19 Epidemiology of Kidney Disease

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CHAPTER OUTLINE

KEY POINTS

- The chronic kidney disease (CKD) definition has remained stable since 2002, with staging now including cause, glomerular filtration rate, and proteinuria categories.
- The prevalence of CKD globally is approximately 10%, with a suggestion of higher and rising prevalence in low- and middle-income countries.
- Risk factors for CKD incidence and progression provide a foundation for prevention.
- Genetic susceptibility to CKD is now better understood, including a large effect by the *APOL1* susceptibility locus in populations of African descent.
- The incidence of end-stage kidney disease (ESKD) has stabilized at a high level in the United States and a number of other high-income countries but enormous racial and ethnic disparities exist globally, with rates increasing in low- and middle-income countries.
- Kidney transplantation provides the best treatment for ESKD and, despite a limited supply of organs, the prevalence of transplant recipients is increasing markedly, forming an important subset of patients with CKD.
- Acute kidney injury is both a consequence of CKD and a cause of further CKD progression.

INTRODUCTION

There has been growing interest in the epidemiology of kidney disease in recent decades, fueled by the rising prevalence of kidney failure and observations of disproportionate disease burden by region, race, and clinical characteristics. Kidney disease epidemiology, the study of disease incidence, distribution, and determinants, has enabled substantial progress in the field. Notable developments over the past decade include the issuance of guidelines for acute and chronic kidney disease (CKD), with consensus definitions based on widely used clinical measures, the effective use of disease registries and electronic health records to identify and address areas of unmet need, the discovery of genetic variants that increase kidney disease risk, and the application of innovative technologies for the discovery of biomarkers of kidney disease.

CONSENSUS DEFINITIONS, CONCEPTUAL MODELS, AND CLASSIFICATION OF CHRONIC AND ACUTE KIDNEY DISEASE

The terms to describe chronic and acute kidney disease have evolved over the years. Consensus definitions were first provided in clinical practice guidelines in the early 2000s, a critical step to allow the clinical translation of epidemiologic research. The definition and classification for CKD was proposed by the U.S. National Kidney Foundation (NKF) Kidney Disease Outcome Quality Initiative (KDOQI) in 2002,[1] adopted by the international guideline group Kidney Disease Improving Global Outcomes (KDIGO) in 2005,[2] and updated by KDIGO in 2012.[3] The definition and classification of acute kidney injury (AKI) were proposed by the Acute Dialysis Quality Initiative (ADQI) in 2004,[4] modified by the Acute Kidney Injury Network (AKIN) in 2007,[5] and harmonized

Table 19.1 Definitions and Classification of Chronic Kidney Disease (CKD) and Acute Kidney Injury (AKI)

Parameter	CKD	AKI
Definition		
Functional criteria	GFR < 60 mL/min/1.73 m^2	Increase in serum creatinine level (SCr) by 50% within 7 days *or* Increase in SCr by 0.3 mg/dL within 2 days, *or* Oliguria (urine output <0.5 mL/kg/h for >6 hours)
Structural criteria	Kidney damage	None
Duration	>3 months	≤1 week
Classification		
Cause	Presence or absence of systemic disease and location in the kidney of pathologic–anatomic abnormalities (e.g., glomerular, tubulointerstitial, vascular, cystic)	Pathophysiology (decreased kidney perfusion, obstruction of the urinary tract, parenchymal kidney disease other than acute tubular necrosis, and acute tubular necrosis)
Severity (stage)	Stages based on GFR (G) and albuminuria (A) categories, and related terms[a]: G1 ≥90: Normal or high G2 60–89: Mildly decreased[b] G3a 45–59: Mildly to moderately decreased G3b 30–44: Moderately to severely decreased G4 15–29: Severely decreased G5 <15 or KRT (kidney failure) A1 <0: Normal to mildly increased A2 30–300: Moderately increased[b] A3 >300: Severely increased[b]	Stages based on SCr or urine output: 1. SCr ≥1.5–1.9 times baseline *or* >0.3-mg/dL increase *or* urine output <0.5 ml/kg/h for 6–12 hours 2. SCr ≥2.0–2.9 times baseline *or* urine output <0.5 mL/kg/h for ≥12 hours 3. SCr ≥3.0 times baseline *or* to ≥4.0 mg/dL *or* KRT In patients aged <18 years, decrease in eGFR to <35 mL/min/1.73 m^2 *or* urine output <0.3 mL/kg/h for ≥24 hours *or* anuria for ≥12 hours
Risk categories	Moderate (G1–G2, A2; G3a, A1) High (G1–G2, A3; G3a, A2; G3b, A1) Very high (G3a, A3; G3b, A2–A3; G4, A1–A3; G5, A1–A3)	Not specified separately
Treatment	KRT included in stage 5, defined as ESKD	KRT included in stage 3

[a]Units for GFR, mL/min/1.73 m^2; units for albuminuria categories, AER mg/day, ACR mg/g. In the absence of evidence of kidney damage, GFR category G1 or G2 does not fulfill the criteria for CKD. In the absence of GFR < 60, albuminuria category A1 does not fulfill the criteria for CKD.

[b]Terms for categories G2 and A2 are relative to young adult levels. Category A3 includes nephrotic syndrome (albumin excretion usually >2200 mg/day [ACR >2220 mg/g; >220 mg/mmol]).

ACR, Albumin-to-creatinine ratio; *AER,* albumin excretion rate; *CKD,* chronic kidney disease; *ESKD,* end-stage kidney disease; *GFR,* glomerular filtration rate; *KRT,* kidney replacement therapy; *SCr,* serum creatinine concentration.

Adapted from Kidney Disease: Improving the Improving Global Outcomes (KDIGO) Acute Kidney Work Group. KDIGO clinical practice guideline for acute kidney injury. *Kidney Int Suppl.* 2012,2:1–138; and Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl.* 2013;3:1–150.

by KDIGO in 2011, which also provided a definition for acute kidney diseases and disorders (AKDs).[6] The definitions and staging systems for CKD and AKI are based on kidney measures that are frequently obtained in clinical practice and population studies (Table 19.1).

The conceptual models for the development, progression, and complications of chronic and acute kidney diseases are similar.[3,6] They include antecedents associated with an increased risk for the development of kidney disease, stages of disease, and complications of disease, including death (Fig. 19.1). Risks for the development of kidney disease may be categorized as susceptibility due to demographic and genetic factors or exposure to factors that might initiate kidney disease. Kidney disease is defined by abnormalities in structure or function, with abnormalities in kidney structure (damage) often, but not always, preceding abnormalities in function. Markers of kidney damage include pathologic findings on biopsy, abnormalities in urine sediment or imaging, proteinuria, and alterations in tubular function. Kidney function is best measured by the glomerular filtration rate (GFR). A GFR < 60 mL/min/1.73 m^2 is considered moderately decreased kidney function, and a GFR < 15 mL/min/1.73 m^2 is defined as kidney failure. Complications include conditions that occur more often in people with compared with people without kidney disease. In addition to metabolic and hormonal disorders associated with uremia (e.g., anemia, mineral and bone disease, malnutrition, neuropathy), complications include cardiovascular disease (CVD), drug toxicity, and a wide variety of other conditions, such as infections, cognitive impairment, and frailty. Death may result from kidney failure or complications.

Kidney disease is classified according to its duration, severity, cause, and prognosis (Fig. 19.2).[3] Duration of kidney disease longer than 3 months is defined as CKD, whereas duration less than or equal to 3 months is defined as AKD, of which AKI is the subset of cases that occur within 7 days. Criteria

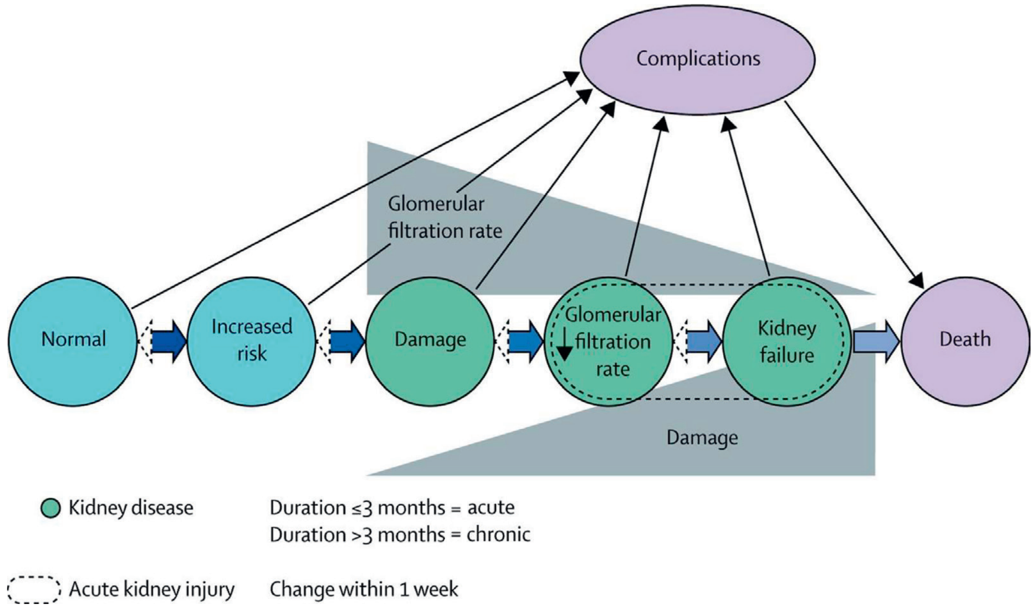


Fig. 19.1 Factors associated with increased risk of kidney disease *(blue),* stages of disease *(green),* and complications (including death; *purple*). (From Eckardt KU, Coresh J, Devuyst O, et al. Evolving importance of kidney disease: from subspecialty to global health burden. *Lancet*. 2013;382[9887]:158–169.)

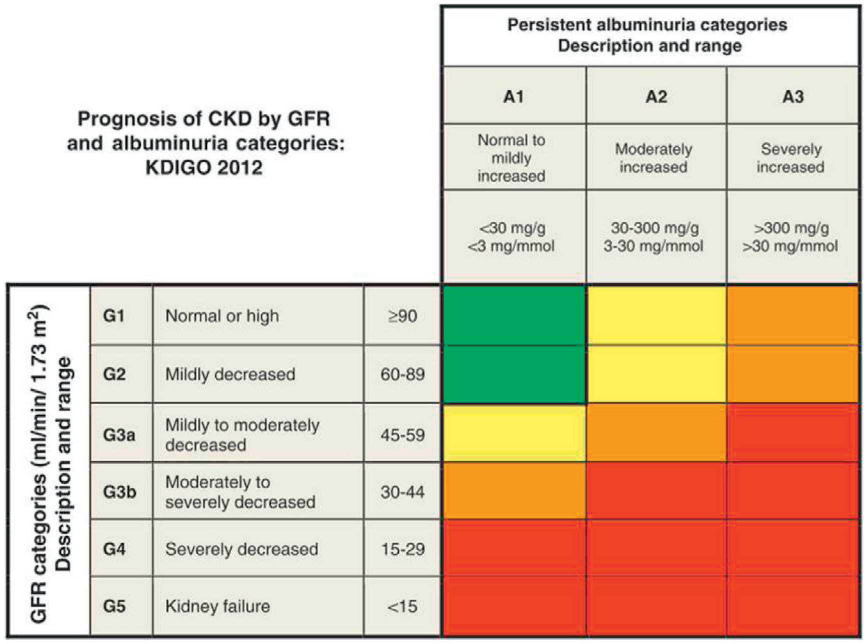


Fig. 19.2 Chronic kidney disease staging. Classification is based on the identification of cause (C), glomerular filtration rate *(GFR)* category *(G)*, and albuminuria *(A)*. This is collectively known as "CGA staging." (From Kidney Disease: Improving Global Outcomes [KDIGO] CKD Work Group. Clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl*. 2013;3:1–150.)

for CKD include GFR < 60 mL/min/1.73 m^2 or markers of kidney damage. The severity of CKD is assessed by the level of GFR (stages G1–G5) and albuminuria (stages A1–A3), with end-stage kidney disease (ESKD) as the subset of stage G5 treated by dialysis and transplantation, collectively known as "kidney replacement therapy" (KRT). Causes of CKD are classified according to the presence or absence of systemic disease and location of pathologic and anatomic abnormalities within the kidney. Prognosis is determined based on G and A staging and categorized as moderate, high, and very high risk.

Criteria for AKI include a rise in the serum creatinine level or oliguria, and severity is staged according to peak value for the serum creatinine level or nadir value for urine output. Patients requiring acute KRT are classified as stage 3, the most severe stage. Causes of AKI are classified by pathophysiology as decreased kidney perfusion, obstruction of the urinary tract, and parenchymal kidney diseases (including acute tubular necrosis); all may occur in people with or without underlying CKD. Incomplete recovery of AKD can lead to new-onset CKD in those without underlying CKD or progression of CKD in those with underlying CKD. In addition, there is overlap among the risk factors, causes, and complications between AKD and CKD.

PRINCIPLES OF EPIDEMIOLOGY AND USE IN KIDNEY DISEASE RESEARCH

This chapter focuses on the prevalence, incidence, outcomes, and risk factors for kidney disease. Disease prevalence is the proportion of individuals in a given population with disease; disease incidence is the number of new cases that develop among individuals without prevalent disease over a given period of time. Disease prevalence thus depends not only on disease incidence, but also on survival and how long the disease persists. Outcomes refer to events of interest that may occur with greater frequency among affected individuals, and risk factors refer to characteristics that precede the onset of disease or outcome. It should be noted that much of the epidemiologic literature is observational, and thus observed associations among risk factors, kidney disease, and outcomes may be causal or noncausal. The Bradford Hill criteria provide guidelines for inferring causation and specify the minimal conditions that need to be met to support a causal relationship between the risk factor (exposure) and disease (outcome; Table 19.2).[7] Trends refer to changes in disease incidence, prevalence, or outcomes over time and may be due to changes in population demographics, risk factor distribution, risk factor control, or availability and effectiveness of treatment. The latter explanation is particularly relevant for trends in ESKD, which is defined by treatment with KRT.

Table 19.2 Bradford Hill Criteria for Causation

Criteria	Explanation
Strength of association	The stronger the association, the more likely the relationship is causal.
Consistency	A casual association is consistent when replicated in different populations and studies.
Specificity	A single putative cause produces a single effect.
Temporality	Exposure precedes outcome; that is, risk factor predates disease.
Biologic gradient	Increasing exposure to a risk factor increases risk of the disease, and reduction in exposure reduces risk.
Plausibility	The observed association is consistent with biologic mechanisms of disease processes.
Coherence	The observed association is compatible with existing theory and knowledge within a given field.
Experimental evidence	The factor under investigation is amenable to modification by an appropriate experimental approach.
Analogy	An established cause and effect relationship exists for a similar exposure or disease.

Adapted from Hill AB. The environment and disease: association or causation? *Proc R Soc Med.* 1965;58:295–300.

RECENT CONTRIBUTIONS OF EPIDEMIOLOGY TO NEPHROLOGY

Epidemiologic studies underlie many of the important advances in kidney disease in the past 2 decades. The development of GFR-estimating equations and recognition of albuminuria as a risk factor for adverse outcomes were central to the definition and staging of CKD, which in turn enabled studies of the prevalence of CKD, identification of CKD as a risk factor for CVD, and recognition of CKD as a global public health problem. In addition, estimated GFR (eGFR) and albuminuria are now widely used as prognostic markers in clinical practice and as surrogate markers of kidney disease outcomes in clinical trials of treatment for kidney disease.[8] Advances in genetic epidemiology have revolutionized our understanding of both common and uncommon diseases.[9] Common variations among individuals of African ancestry in a single gene encoding apolipoprotein L1 (APOL1) are now recognized to increase the risk for ESKD.[10–13] Similarly, large cohort studies have demonstrated that the presence of sickle cell trait increases the risk for both CKD and ESKD, increasing our understanding of racial disparities in kidney disease.[14,15] Genome-wide association studies in membranous and immunoglobulin A (IgA) nephropathy have provided strong evidence of heritability and a role for the immune system in the development of disease.[16–20] Rare genetic variants have been identified as the cause of several monogenic diseases, such as polycystic kidney disease, familial nephrotic syndrome, and congenital anomalies of the kidney and urinary tract (CAKUT).[21,22] Global collaboration, coupled with the increasing availability of large databases with rigorously defined clinical characteristics, frequent ascertainment of kidney measures, and genomic, proteomic, and metabolomic data will allow continued advances in research, clinical practice, and public health.

CHRONIC KIDNEY DISEASE

EPIDEMIOLOGY

Obtaining accurate and comparable estimates of CKD burden across world regions is difficult for several reasons. Identification of disease requires testing for kidney disease measures (e.g., serum creatinine or cystatin C for eGFR, urine albumin, urine total protein), which are not uniformly performed. Differences across populations may reflect differences in laboratory testing practice, because CKD in its early stages is often asymptomatic, with low awareness of disease.[23,24] For example, if only eGFR is ascertained, prevalence estimates reflect CKD stages G3–G5 and will be lower than if albuminuria is also ascertained, enabling prevalence estimates of CKD stages G1–G5 and A1–A3. Two populations with equal proportions of CKD also may have very different prevalence estimates if one performs near-universal laboratory testing and the other screens only high-risk individuals. In addition, because CKD is defined using kidney disease measures, assays must be standardized to derive comparable estimates. In the United States, standardization of creatinine assays was achieved in the past decade[25]; however, standardization of cystatin C and urine albumin assays are still ongoing. Finally, estimating equations for GFR are not consistently applied, and there is known variation in CKD prevalence when, for example, the Modification of Diet in Renal Disease (MDRD) or Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equations are used.[26,27] Progress in CKD awareness, testing, definitions, and measurement should enable substantial progress in the field in the coming decade.

PREVALENCE

Estimates of the prevalence of CKD worldwide vary, but generally range between 8% and 16%.[28] The most accurate CKD prevalence estimates come from nationally representative surveys, such as the National Health and Nutritional Examination Survey (NHANES) in the United States. In this sampling design, people are selected from the general population and undergo detailed physical and laboratory examinations. Estimates of disease prevalence are obtained by weighting sample proportions by inverse probability of selection from the general population. In NHANES 2011–2012, the estimate of U.S. prevalence of CKD was based on a one-time measurement of eGFR creatinine (Cr) and the urine albumin-to-creatinine (ACR) ratio, was 14.2%, representing 7.3% for stages G1 and G2 and 6.9% for stage G3 or G4.[29] It should be noted that this study did not assess the chronicity of abnormalities of these measures and thus may have slightly overestimated their prevalence. However, other considerations suggested an underestimation of prevalence of CKD. Markers of kidney damage other than albuminuria were not assessed. People with CKD, stage G5—those undergoing KRT and those with an eGFR < 15 mL/min/1.73 m^2—were not included in the NHANES estimates. In addition, NHANES does not include people living in institutionalized settings or who are involved in the military.

Despite the relatively high prevalence of CKD in the United States, the recent NHANES study did report some encouraging trends.[29] An earlier study of NHANES showed an increasing prevalence of CKD over time (from 10.0%–13.1% from 1988–1994 to 1999–2004, using a definition that adjusted for short-term variability in the urine ACR).[30] Since that time, however, a recent study has suggested that the prevalence of the CKD remained unchanged (14.0% in 2003–2004 to 14.2% in 2011–2012), which is all the more encouraging when one considers the significant increases in the prevalence of diabetes mellitus and obesity.[31,32] Similarly, the prevalence of CKD stages G3 and G4 was stable when participants were stratified by age, gender, and diabetes status (Fig. 19.3). In

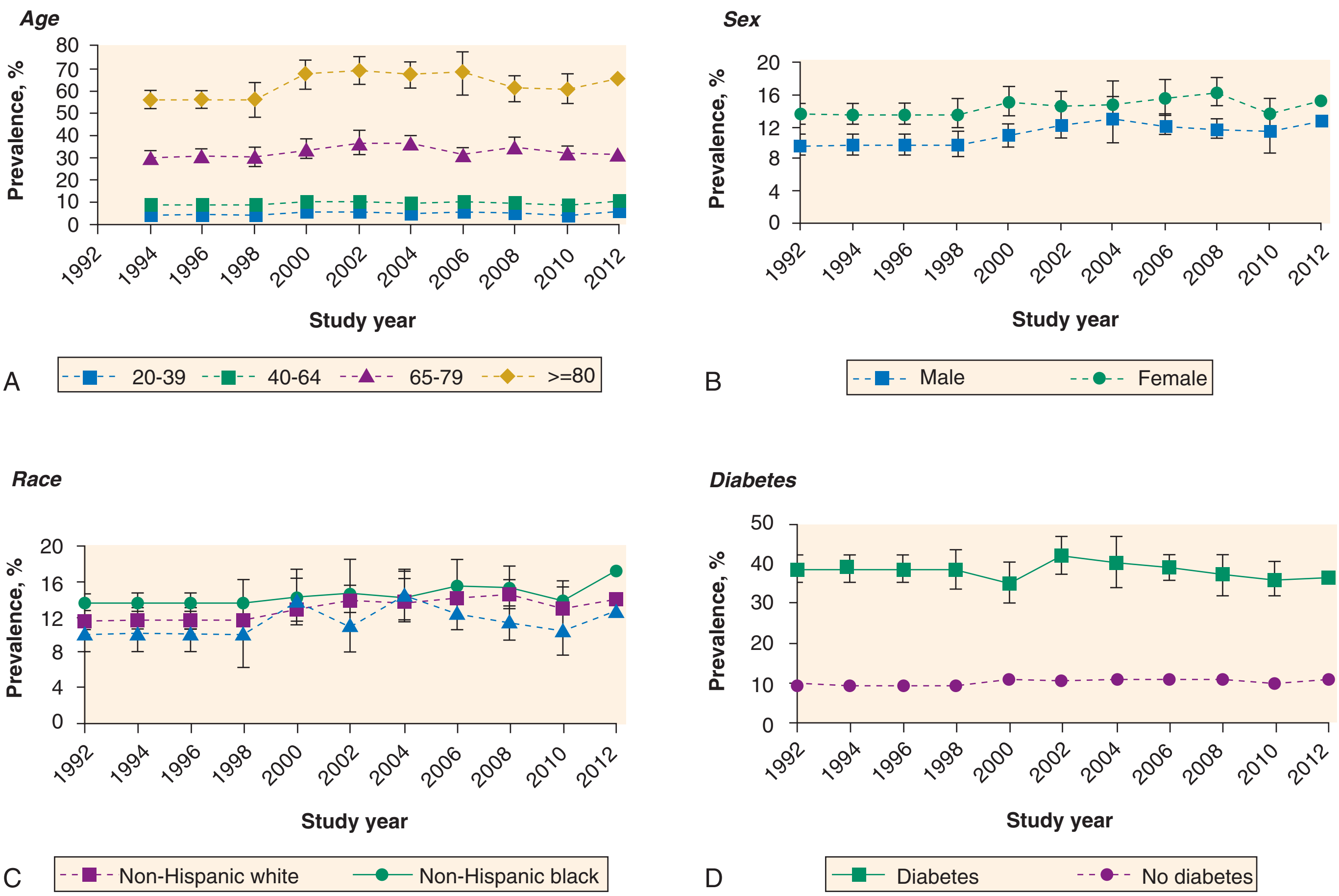


Fig. 19.3 Adjusted prevalence of chronic kidney disease (CKD) stages 3 and 4 (estimated glomerular filtration rate of 15–59 mL/min/1.73 m^2, calculated with the Chronic Kidney Disease Epidemiology Collaboration equation [CKD-EPI]; see text[26,27]) in U.S. adults by age (A), sex (B), race and ethnicity (C), and presence or absence of diabetes (D), NHANES 1988–1994 through 2011–2012. Each subgroup is adjusted for the other three subgroup variables (e.g., age-specific prevalence is adjusted for gender, race and ethnicity, and diabetes status). *NHANES,* National Health and Nutrition Examination Survey. (Adapted from Murphy D, McCulloch CE, Lin F, et al. Trends in prevalence of chronic kidney disease in the United States. *Ann Intern Med.* 2016;165:473–481.)

fact, the only subgroup tested in which crude prevalence meaningfully increased from 2003–2004 to 2011–2012 was non-Hispanic blacks (4.9% in 2003–2004 to 6.2% in 2011–2012). Given the already disproportionate burden of CKD among black people, these results are worrisome, and additional efforts are needed to address racial disparities in CKD. Subsequent phases of NHANES will confirm whether these trends persist.

In Europe, no multinational survey data exist, so a study combined data from 19 general population studies in 13 European countries to estimate age- and gender-adjusted CKD prevalence.[33] There was large heterogeneity by country, even when stratified by diabetes, hypertension, and obesity status, with lower rates of CKD stages G3–G5 in central Italy (1%) but higher in northeast Germany (5.9%). Similarly, the prevalence of CKD stages G1–G5 ranged from 3.3% in Norway to 17.3% in northeast Germany. Fig. 19.4 indicates the differences by region in the population aged 45 to 74 years, and highlights the relatively localized source populations, which may not be representative of all of Europe. Other limitations noted by the authors included the heterogeneity in laboratory methods, with not all assays being isotope dilution mass spectrometry (IDMS)-standardized, and differences in dietary habits, with certain high-protein diets typical of northern Germany known to influence serum creatinine (Cr), resulting in an underestimate of eGFRcr.

Global estimates of CKD are also heterogeneous (Table 19.3; see also Chapter 75).[34,35] A systematic analysis of 33 studies from 32 countries (nearly 50% of the world's population) has found the crude prevalence of CKD stages G1–G5 ranges in men from 4.5% in South Korea to 25.7% in El Salvador, and in women from 4.1% in Saudi Arabia to 16.0% in Singapore.[34] In addition to 17 studies that were included which were derived from a regional or multisite source population and thus not necessarily generalizable to the nation as a whole, underlying studies differed in the creatinine measurement method, the eGFR Cr equation used, and the protocol for assessing albuminuria. Noting these caveats, the authors estimated the age-standardized global prevalence of CKD in 2010 as 10.4% in men and 11.8% in women. They also reported substantial differences by gross national product, with high-income countries estimated to have lower CKD prevalence (8.6% and 9.6% in men and women, respectively) than low- and middle-income countries (10.6% and 12.5%).

Diabetes and hypertension are considered the major causes of CKD globally.[28] In the U.S. population with diabetes mellitus, an estimated 39.4% had CKD in 2011–2014; in the population with hypertension, an estimated 32.1% had CKD, and in the population with obesity, an estimated 17.6% had CKD.[24] Interestingly, measures of kidney disease varied slightly within each comorbidity—47% of the CKD within the diabetes group was manifest by albuminuria only, 27% by decreased GFR only, and 26% by both, compared with 36%, 45%, and 19%, respectively, for hypertension and 30%, 55%, and 15%, respectively, for obesity (Fig. 19.5). The prevalence of CKD by specific cause is difficult to estimate, but if the prevalence of diabetes mellitus in the United States is approximately 10%,[34] and over one-third of people with diabetes mellitus have CKD,[36] then the prevalence of diabetic kidney disease in the U.S. population may be close to 3% to 4%, or one-third of all prevalent CKD cases. In contrast to trends in the prevalence of kidney disease in those with diabetes mellitus, which has been estimated to be fairly flat,[29] trends in the prevalence of diabetic kidney disease in the general population have been reported to follow the increasing trends in the prevalence of diabetes.[37]

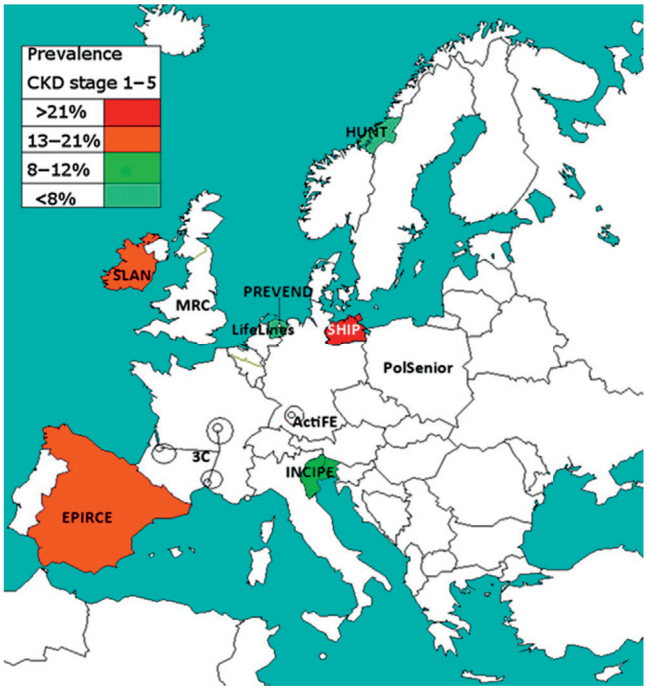


Fig. 19.4 Adjusted prevalence of chronic kidney disease (CKD) stages 1–5 in the population aged 45–74 years, in isotope dilution mass spectrometry (IDMS) studies. Prevalence was age- and gender-adjusted to the EU27 population of 2005. The study names in the uncolored regions are studies that used non–IDMS-standardized creatinine or studies that recruited subjects aged 50+ years. *IDMS,* Isotope dilution mass spectrometry. (From Brück K, VS, Gambaro G, et al. CKD prevalence varies across the European general population. *J Am Soc Nephrol.* 2016;27:2135–2147.)

INCIDENCE

Compared with estimates of CKD prevalence, information on the incidence of CKD is relatively sparse. Calculation of the incidence—the development of new cases of CKD—requires a population-representative sample, longitudinal follow-up, and regular and comparable ascertainment of kidney measures in each individual. The incidence of disease stemming from clinical populations may not be valid, because higher-risk individuals are more likely to receive testing. On the other hand, the incidence of disease stemming from population-based cohorts may underestimate CKD incidence due to loss to follow-up and relatively infrequent study visits (e.g., visits every 3 or 5 years).

In the United States, a few long-running prospective cohorts have provided estimates of incident CKD, but the frequency of assessment of kidney measures differed. The Atherosclerosis Risk in Communities (ARIC) study, a cohort of middle-aged black and white individuals aged 45 to 65 years of age, from 4 U.S. communities, estimated an incidence of 10.4 cases of CKD stages G3–G5/1000 years of follow-up, or a 7.3% incidence over 8.8 years when eGFRcr was assessed every 3 years.[38] The Framingham study, a community-based study of

Table 19.3 Global Estimates of Chronic Kidney Disease[a]

	High-Income Countries				Low- and Middle-Income Countries			
	CKD Stages 1–5		CKD Stages 3–5		CKD Stages 1–5		CKD Stages 3–5	
Age (Years)	*Men*	*Women*	*Men*	*Women*	*Men*	*Women*	*Men*	*Women*
Prevalence, % (95% CI)								
20–29	3.7 (2.7–5.1)	5.3 (3.8–6.3)	0.7 (0.3–1.4)	0.9 (0.4–1.6)	7.3 (6.4–9.0)	6.6 (6.2–7.3)	3.0 (1.7–5.4)	2.0 (1.4–3.3)
30–39	5.0 (4.0–6.0)	5.9 (4.4–6.9)	1.3 (0.7–2.1)	1.6 (0.9–2.6)	8.1 (6.8–10.3)	9.0 (8.6–9.7)	3.1 (1.9–5.0)	3.1 (2.1–5.1)
40–49	6.8 (5.5–8.2)	7.7 (5.9–9.0)	2.1 (1.4–3.1)	3.2 (2.0–4.8)	10.2 (9.0–12.6)	11.5 (11.0–12.7)	3.0 (1.8–6.2)	4.0 (2.5–7.7)
50–59	10.2 (8.6–12.2)	11.1 (8.6–13.8)	4.6 (3.2–6.7)	7.4 (5.2–10.2)	12.0 (10.4–15.1)	15.7 (14.7–17.8)	4.9 (3.5–8.1)	6.7 (4.5–11.7)
60–69	16.0 (13.6–18.1)	15.6 (12.6–18.9)	9.4 (7.8–11.6)	12.2 (9.6–15.8)	16.3 (14.7–20.0)	21.3 (19.6–24.9)	9.7 (6.1–15.6)	13.1 (9.5–19.7)
≥70	28.1 (23.4–33.0)	28.9 (22.7–34.3)	22.7 (20.0–26.4)	28.5 (26.1–31.5)	20.6 (19.4–24.1)	28.4 (26.3–32.7)	11.8 (9.0–17.6)	17.3 (12.6–27.2)
Total	10.1 (8.8–11.1)	12.1 (9.9–13.7)	5.4 (4.6–6.5)	8.6 (6.9–10.7)	10.2 (9.1–12.4)	12.1 (11.6–13.3)	4.3 (2.9–7.1)	5.3 (3.8–8.2)
Age-standardized	8.6 (7.3–9.8)	9.6 (7.7–11.1)	4.3 (3.5–5.2)	5.7 (4.4–7.6)	10.6 (9.4–13.1)	12.5 (11.8–14.0)	4.6 (3.1–7.7)	5.6 (3.9–9.2)
Absolute Numbers (in Thousands; 95% CI)								
20–29	3453 (2485–4718)	4647 (3307–5500)	694 (288–1275)	800 (307–1570)	37,121 (32,209–45,651)	32,054 (30,213–35,932)	14,998 (8471–27,644)	9863 (6746–16,328)
30–39	4699 (3802–5684)	5298 (4000–6235)	1221 (659–1957)	1440 (712–2514)	33,274 (27,740–42,285)	35,920 (34,256–38,727)	12,499 (7803–20,591)	12,265 (8185–20,402)
40–49	6326 (5158–7691)	7120 (5519–8386)	2004 (1290–2934)	2990 (1731–4700)	35,322 (31,190–43,673)	39,012 (37,165–43,201)	10,253 (6328–21,335)	13,495 (8579–26,111)
50–59	8531 (7149–10,157)	9752 (7541–12,136)	3874 (2625–5549)	6485 (4218–9418)	29,618 (25,640–37,287)	38,337 (35,959–43,651)	12,167 (8612–19,988)	16,355 (10,900–28,611)
60–69	9579 (8155–10,849)	10,425 (8411–12,690)	5608 (4694–6963)	8291 (6329–11,264)	22,434 (20,338–27,637)	31,136 (28,731–36,396)	13,420 (8430–21,497)	19,232 (13,883–28,848)
≥ 70	15,699 (13,066–18,416)	24,426 (19,150–29,003)	12,690 (11,136–14,745)	24,065 (21,769–27,161)	19,606 (18,459–22,925)	33,637 (31,150–38,706)	11,176 (8541–16,673)	20,509 (14,912–32,240)
Total	48,285 (42,349–53,284)	61,669 (50,394–69,929)	26,091 (22,014–31,078)	44,071 (35,296–54,831)	177,375 (159,157–215,924)	210,096 (200,787–231,704)	74,513 (50,324–123,460)	91,719 (66,697–143,121)

[a]Age-specific and age-standardized prevalence estimates and absolute numbers of men and women with CKD in high-, low-, and middle-income countries.

CI, Confidence interval; *CKD,* chronic kidney disease.

From Mills KT, Xu Y, Zhang W, et al. A systematic analysis of worldwide population-based data on the global burden of chronic kidney disease in 2010. *Kidney Int.* 2015;88(5):950–957.

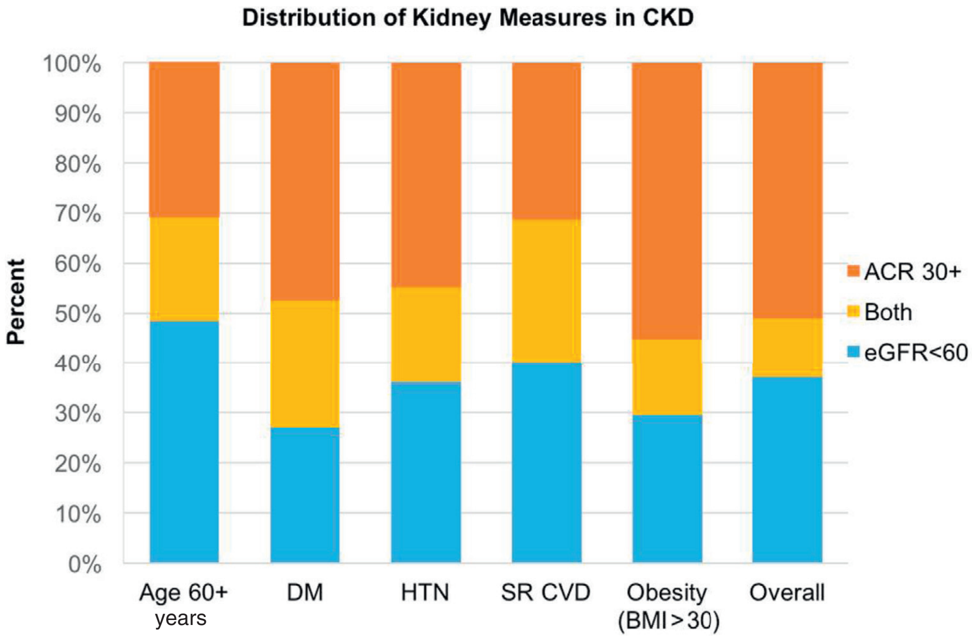


Fig. 19.5 Distribution of kidney measures in NHANES participants with chronic kidney disease (CKD) older than 60 years and those with diabetes mellitus *(DM)*, hypertension *(HTN)*, self-reported cardiovascular disease *(SR CVD)*, obesity (BMI > 30), and overall. Single-sample estimates of eGFR and ACR; eGFR calculated using the Chronic Kidney Disease Epidemiology Collaboration equation (CKD-EPI) equation. *ACR,* Urine albumin-to-creatinine ratio; *BMI,* body mass index; *eGFR,* estimated glomerular filtration rate; *NHANES,* National Health and Nutrition Examination Survey. (Adapted from United States Renal Data System. 2016 USRDS annual data report: epidemiology of kidney disease in the United States. https://www.ajkd.org/article/S0272-6386(16)30703-X/fulltext.)

predominantly white individuals with a relatively low rate of diabetes (2.7% vs. 11.4% in the ARIC study), and a mean age of 43 years, reported 9.4% of 2585 individuals developed CKD stages G3–G5 in a subsequent visit 18.5 years later.[39] In the Cardiovascular Health Study, a community-based longitudinal study of older adults with a mean age of 72 years, 10% developed CKD stages G3–G5 by year 7 when assessed using eGFRVt, and 19% developed CKD stages G3–G5 when assessed using eGFRcys.[40] In each of these studies, the rate of incident CKD was higher with older age. Notably, none assessed the incidence of albuminuria, so they likely underestimated the incidence of CKD.

Racial differences in incident CKD also exist. In the Coronary Artery Risk Development in Young Adults cohort, a study of black and white individuals aged 18–30 years, the risk of developing CKD stages G3–G5 (in this study, assessed from eGFRcr and requiring an accompanied 25% decline in eGFR) at 10, 15, or 20 years after baseline was 2.6 times higher in blacks than whites.[41] Similarly, there were large racial disparities in the Multi-Ethnic Study of Arteriosclerosis (mean age, 60 years), in which blacks had more than threefold higher risk of developing CKD stages G3–G5 (in this study, assessed from eGFRcys and required a ≥1 mL/min/1.73 m^2 decline in eGFR) compared with whites in a population with baseline eGFR > 90 mL/min/1.73 m^2 (3.4 cases/1000 years vs. 0.9 cases/1000 years).[42] Incidence rates were higher among participants with eGFR between 60 and 90 mL/min/1.73 m^2, and racial differences persisted, albeit slightly attenuated (26.3 cases/1000 years vs. 18.5 cases/1000 years in blacks and whites, respectively). When modeled as incidence over a lifetime using NHANES and registry data, racial disparities again were present, with African Americans projected to develop CKD stages G3–G5 at younger ages and greater disparities in the incidence of more severe disease (Fig. 19.6).[43]

In Europe, the incidence of CKD assessed using eGFRcr and urine albumin excretion >30 mg/day has been reported in the PREVEND cohort, a population-based cohort that oversampled for people with urine albumin concentrations ≥10 mg/L on a first morning void urine sample.[44] Participants were followed over 10 years at 5 separate visits. The incidence of developing CKD stages G1–G5 was 18.6/1000 person-years; the incidence of developing albuminuria was 13.4/1000 person-years, and the incidence of developing decreased GFR was 5.8/1000 person-years. In an administrative cohort, the incidence of serum creatinine ≥1.7 mg/dL (eGFR ranging from 37–57 mL/min/1.73 m^2 for men and 28–43 mL/min/1.73 m^2 for women) was 1701 cases/million population in Southampton, United Kingdom, with much higher rates of CKD among older compared with younger inhabitants.[45]

OUTCOMES

CKD has been increasingly recognized as a top public health priority due to its strong association with adverse outcomes. In some areas of the world, CKD is one of the top five causes of death.[46] Lower GFR and higher albuminuria levels result in a significantly diminished life expectancy at all ages (Fig. 19.7).[24] Part of this increase in mortality is exerted through the higher risk of kidney failure, both untreated and treated by dialysis or transplantation.[47] In addition, earlier stages of CKD are associated with increased risk of death, AKI, CKD progression, CVD, heart failure, and even some types of cancer.[48–52] For AKI as well as other adverse outcomes, risk

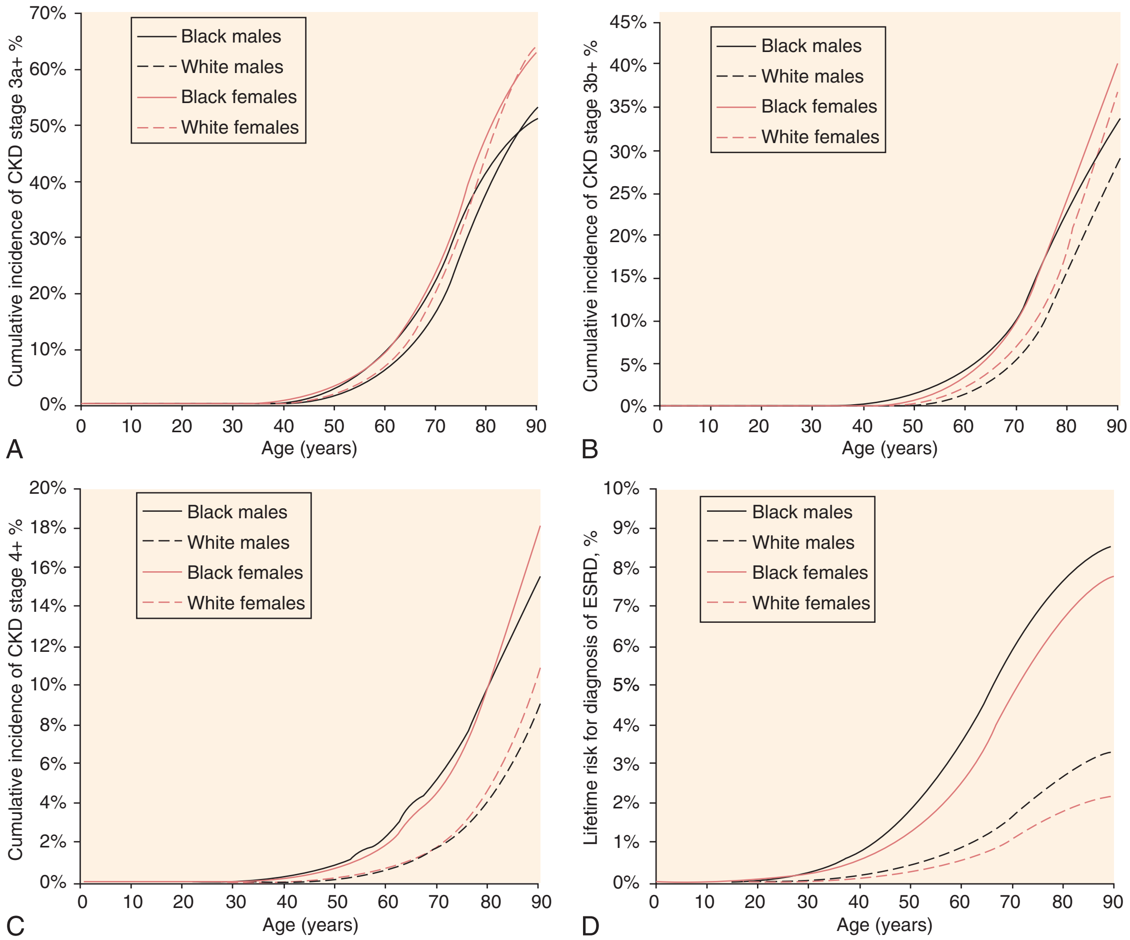


Fig. 19.6 Cumulative incidence from birth by race and gender of chronic kidney disease *(CKD)* stage 3a (A), 3b (B), 4 (C), and end-stage renal disease (*ESKD;* D). (From Grams ME, Chow EK, Segev DL, et al. Lifetime incidence of CKD stages 3–5 in the United States. *Am J Kidney Dis.* 2013;62:245–252.)

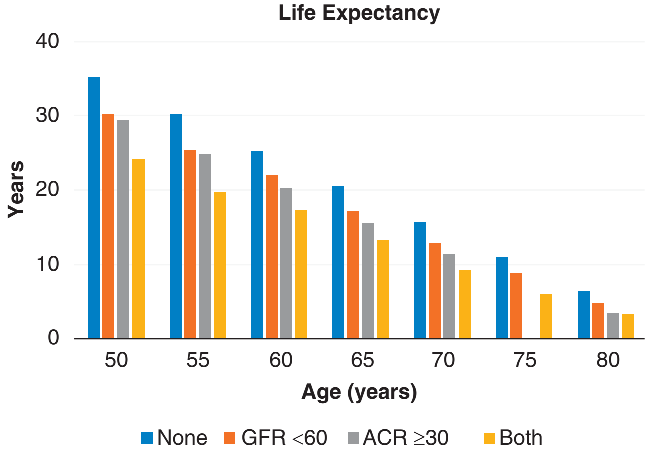
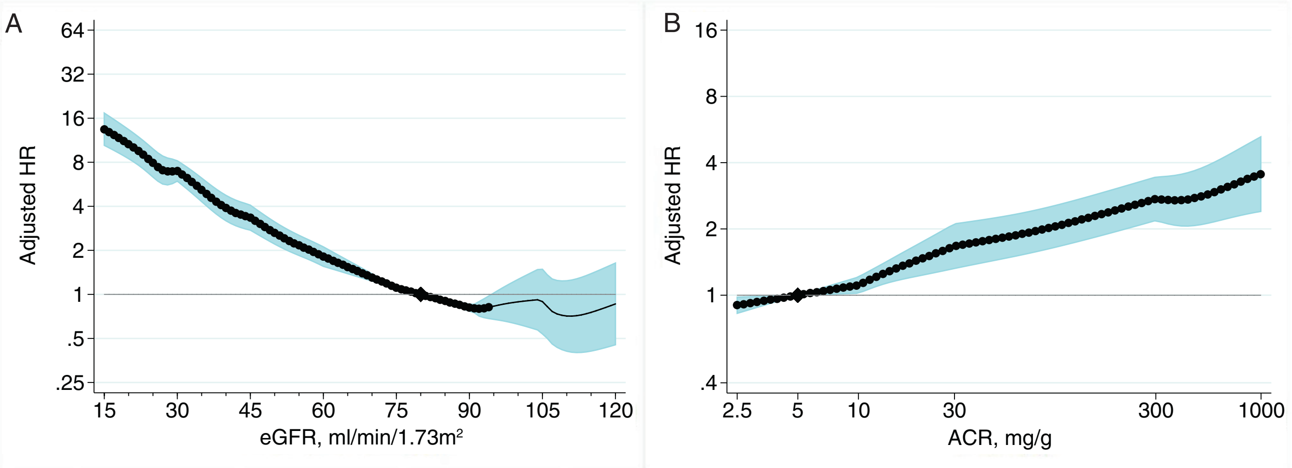


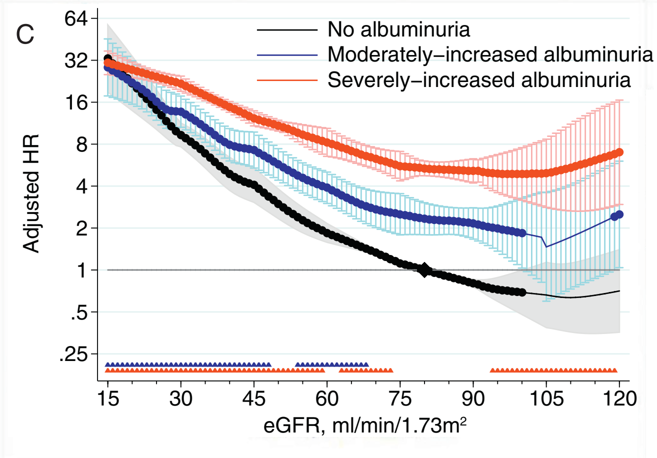
Fig. 19.7 Life expectancy of NHANES participants with or without abnormalities in chronic kidney disease (CKD) measures, 1999–2011. GFR < 60 mL/min/1.73 m^2 or ACR ≥30 mg/g indicate CKD. ACR, urine albumin-to-creatinine ratio; GFR, estimated glomerular filtration rate; *NHANES,* National Health and Nutrition Examination Survey. (Adapted from U.S. Renal Data System. 2016 USRDS annual data report: epidemiology of kidney disease in the united states. https://www.ajkd.org/article/S0272-6386(16)30703-X/fulltext.)

increases with lower GFR and higher albuminuria are generally independent, with each parameter carrying risk, and higher risk with higher stages of disease (Fig. 19.8).[53,54] Risk relationships have been observed in low-risk as well as high-risk groups for ESKD and mortality.[55–58]

There is intense interest in defining earlier outcomes of CKD progression, with the goal of enhancing drug development and clinical prevention.[59–61] Provided that there are no acute effects on eGFR, decline in eGFR by 30% to 40% is now accepted by the U.S. Food and Drug Administration (FDA) as a surrogate outcome for CKD progression, previously assessed by doubling of the serum creatinine level (equivalent to a 57% decline in eGFR) or the development of ESKD.[8] Changes in albuminuria are also associated with ESKD and death,[62] and remission of the nephrotic syndrome is used as a surrogate endpoint in trials of some glomerular diseases.

Kidney measures are not the only risk factors for adverse outcomes, and there is a growing literature on the use of other factors to explain variations in the risk of progression to ESKD within stages of CKD.[63–65] For a given level of GFR or albuminuria, older adults are less likely to experience ESKD than their younger counterparts, possibly due to treatment preference (e.g., refusal of dialysis or transplantation as a therapy) or because of the competing event of death





D

eGFR	ACR/Dipstick <10 / Dip "-"	10-29 / Dip "±"	30-299 / Dip "1+"	300+ / Dip "≥2+"
>105	0.99 (0.59, 1.65)	1.87 (0.99, 3.52)	**3.58** (1.64, 7.82)	**10.36** (4.99, 21.50)
90-104	0.93 (0.74, 1.18)	1.47 (0.95, 2.27)	**2.70** (1.91, 3.82)	**6.86** (4.60, 10.23)
75-89	Ref	1.82 (1.55, 2.14)	**2.90** (2.37, 3.55)	**6.05** (4.70, 7.78)
60-74	**1.57** (1.47, 1.68)	**2.43** (1.89, 3.14)	**3.52** (2.68, 4.62)	**7.95** (6.15, 10.28)
45-59	**2.96** (2.40, 3.65)	**4.49** (3.44, 5.87)	**5.95** (4.49, 7.89)	**10.36** (7.10, 15.13)
30-44	**6.70** (5.54, 8.11)	**7.63** (5.63, 10.34)	**10.85** (8.32, 14.15)	**19.42** (14.16, 26.63)
15-29	**14.72** (13.18, 16.45)	**20.42** (11.31, 36.86)	**19.65** (14.26, 27.09)	**34.17** (21.93, 53.27)

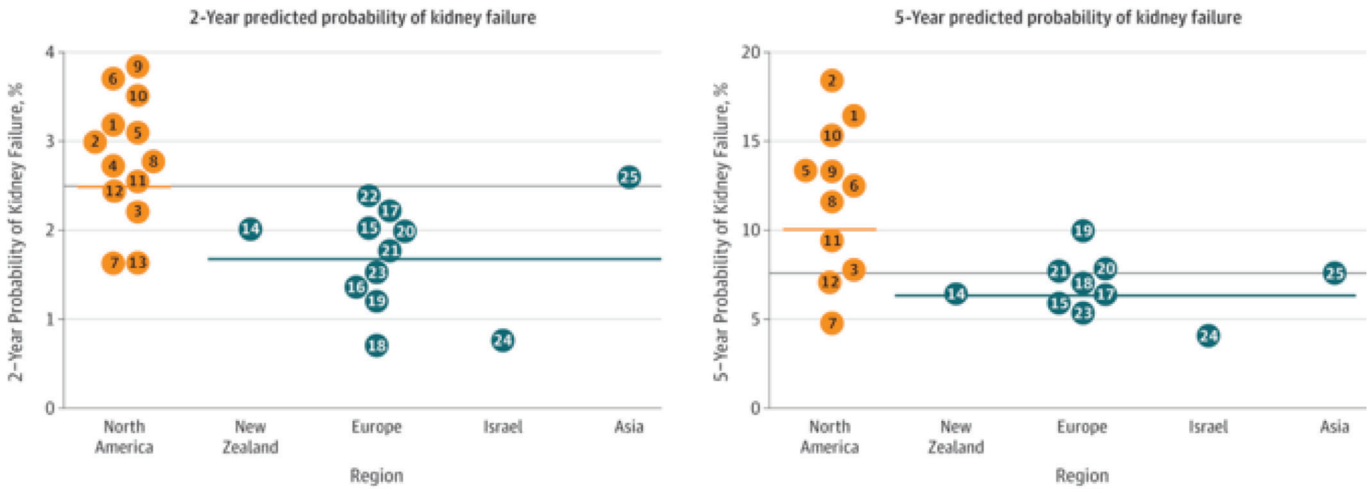
Fig. 19.8 Adjusted hazard ratios *(HRs)* of acute kidney injury in the general population cohorts by level of estimated glomerular filtration rate *(eGFR)* and albuminuria in continuous (A, B, C) and categoric (D) analysis. (C) *Thick lines* indicate statistical significance compared with the reference (*black diamond*) at eGFR of 80 mL/min/1.73 m² in the no-albuminuria group, defined as urine albumin-to-creatinine ratio *(ACR)* <30 mg/g or urine protein dipstick <1+. Stars along the *x*-axis represent significant pointwise directions: the relative risks associated with a particular category of albuminuria compared with the no-albuminuria category at that value of eGFR is significantly different than the corresponding relative risk seen at eGFR of 90 mL/min/1.73 m². A graph without stars would reflect parallel risk associations. HRs were derived from meta-analyses of the general population cohorts and are adjusted for gender, race, body mass index, systolic blood pressure, total cholesterol, history of cardiovascular disease, diabetes, and smoking status. The table represents adjusted HRs derived from categoric analyses of the general population cohorts, with *bold font* representing statistical significance and color coding by risk quartile. (From Grams ME, Sang Y, Ballew SH, et al. A meta-analysis of the association of estimated GFR, albuminuria, age, race, and sex with acute kidney injury. *Am J Kidney Dis.* 2015;66:591–601.)

(i.e., older adults with a given level of GFR or albuminuria are relatively more likely to die than progress to ESKD).[66] Moreover, the absolute risks of ESKD and death may vary based on indications for selection into a study population; research study populations often report higher risk of ESKD than death, whereas clinical population information drawn from medical records data often show a higher risk of death than ESKD.[67,68] Regional variation in the absolute risk of progression to ESKD has also been reported. A validated risk calculator designed to estimate the probability of ESKD in the subsequent 2 to 5 years among patients with CKD stages G3–G5 has incorporated a regional calibration factor to account for the fact that adjusted ESKD risk was higher in North America compared with other countries (Fig. 19.9).[64] Some of this risk differential may be due to the resources allotted to providing treatment rather than underlying characteristics of the countries themselves.[69]

A lower GFR has also been associated with concomitant bone, metabolic, endocrine, and hematologic abnormalities.[70] Well-known pathophysiologic mechanisms underlie the associations of hyperphosphatemia, hyperparathyroidism, metabolic acidosis, and anemia with a lower GFR.[71] The relationship between these abnormalities and albuminuria is less consistent, but the presence of albuminuria does appear to impart a higher risk of hyperparathyroidism and anemia over and above the GFR level.

RISK FACTORS

Age is one of the strongest risk factors for the development of CKD. The prevalence of CKD stages G1–G5 has been reported as nearly 40% in those older than 70 years and over 50% in those 75 years and older in the United Kingdom.[72] Although some have dismissed age as a risk factor, suggesting



North America

Cohort No.	1	2	3	4	5	6	7	8	9	10	11	12	13
Cohort Name	AASK	ARIC	BC CKD	CCF ACR	CRIC	Geisinger	ICES-KDT	MDRD	Mt Sinai BioMe	Pima	REGARDS	Sunnybrook	VA CKD

Non-North America

Cohort No.	14	15	16	17	18	19	20	21	22	23	24	25
Cohort Name	NZDCS	CRIB	GCKD	GLOMMS-1	HUNT	MASTERPLAN	MMKD	NephroTest	RENAAL	SRR-CKD	Maccabi	Gonryo

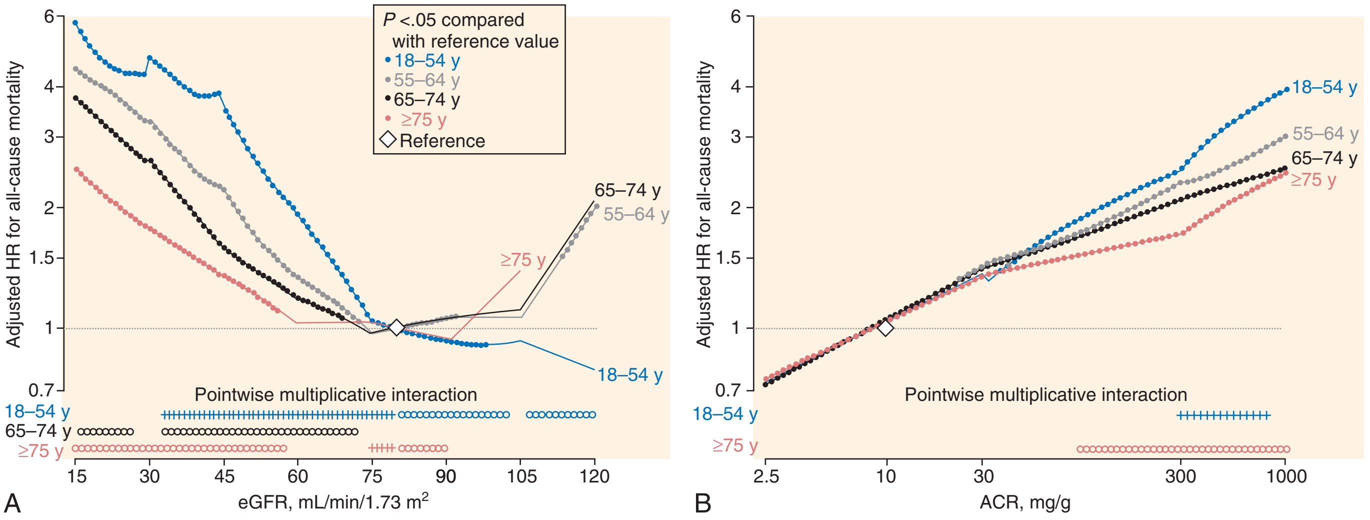
Fig. 19.9 Refit baseline hazard of original four-variable equation at 2 and 5 years in individual cohorts stratified by region. Horizontal *gray line* represents the centered baseline hazard for the original four-variable kidney failure risk equation (age, 70 years; male, 56%; eGFR, 36 mL/min/1.73 m^2; urine albumin-to-creatinine ratio *(ACR)*, 170 mg/g); the *orange* and *blue horizontal lines* represent the weighted mean refit baseline hazard within each region (North America and non–North America). The 25 cohorts included represent studies with available ACR values. Studies with dipstick proteinuria were not included in the calculation. *AASK,* African American Study of Kidney Disease and Hypertension; *ARIC,* Atherosclerosis Risk in Communities; *BC CKD,* British Columbia Chronic Kidney Disease; *CCF,* Cleveland Clinic Foundation; *CRIB,* Chronic Renal Impairment in Birmingham; *CRIC,* Chronic Renal Insufficiency Cohort; eGFR, estimated glomerular filtration rate; GCKD, German CKD; GLOMMS, Grampian Laboratory Outcomes, Morbidity and Mortality Studies; *HUNT,* Nord Trøndelag Health Study; *ICES-KDT,* Institute for Clinical Evaluative Sciences, Provincial Kidney, Dialysis, and Transplantation; *MASTERPLAN,* Multifactorial Approach and Superior Treatment Efficacy in Renal Patients With the Aid of a Nurse Practitioner; *MDRD,* Modification of Diet in Renal Disease; *MMKD*, mild to moderate kidney disease; *NZDCS,* New Zealand Diabetes Cohort Study; *REGARDS,* Reasons for Geographic and Racial Differences in Stroke Study; *RENAAL,* Reduction of Endpoints in Non–insulin Dependent Diabetes Mellitus With the Angiotensin II Antagonist Losartan; *SRR-CKD,* Swedish Renal Registry CKD; *VA CKD,* Veterans Administration CKD. (From Tangri N, Grams ME, Levey AS, et al. Multinational assessment of accuracy of equations for predicting risk of kidney failure: a meta-analysis. *JAMA.* 2016;315:164–174.)

that nephron loss is a normal part of aging,[73,74] associations between a lower GFR and higher albuminuria and adverse outcomes, including both death and ESKD, persisted in all subgroups of age in an individual-level meta-analysis of 2 million participants from 46 cohort studies.[58] On the other hand, among people with stages G3–G5 CKD, older age is negatively associated with risk for ESKD, which may be due to the higher risk of mortality, both overall and attributable to CKD at an older age (Fig. 19.10).[63,64]

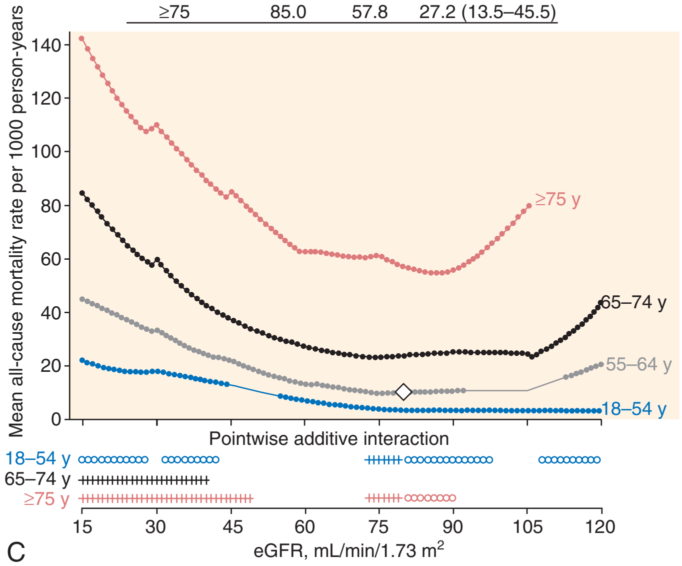
Gender is also a risk factor for CKD, although the relationships are a bit nuanced. Women are at a slightly higher risk for the development of CKD stage G3; however, later stages of CKD are more common in men, suggesting that men may be at higher risk for CKD progression.[43,75] The relationship between eGFR and albuminuria and adverse outcomes was examined by gender in an individual-level meta-analysis of 2 million participants. Men had a higher absolute risk of ESKD, death, and cardiovascular mortality at all levels of eGFR and albuminuria, but the relationship between eGFR and albuminuria and adverse outcomes was similar between the genders.[76]

Compared with Americans of primarily European ancestry, African Americans have a much higher risk of later-stage CKD and CKD progression. These racial differences are only partially explained by differences in socioeconomic or clinical risk factors.[77–80] Interestingly, some studies have suggested that the prevalence and incidence of earlier stages of CKD are similar by race, with only the later stages demonstrating a higher burden among African Americans.[81] For example, in a nationally representative cohort of adults older than 45 years, the prevalence of an eGFR of 50–59 mL/min/1.73 m^2 was 18.9% among black individuals and 31.1% among white individuals, and only at an eGFR < 30 mL/min/1.73 m^2 was there a higher prevalence among blacks compared with whites.[81]

Advances in explaining the racial disparities in CKD include the identification of *APOL1* high-risk alleles, a genetic variant that when present in two copies confers approximately a twofold higher risk of CKD progression and ESKD.[10,12,82,83] The gene variants provide resistance to infection with *Trypanosoma brucei rhodesiense*, a parasite endemic to Africa (Fig. 19.11).[84] The mechanism for increased kidney risk remains uncertain,[85] and *APOL1* high-risk status—present in about 12% of African Americans but less than 1% of whites in population-based studies in the US[86]—explains some but not all of the racial disparities in CKD. The presence of sickle



Age, years	Mean mortality rate/1000 person-years: eGFR, mL/min/1.73 m² 45	80	Difference (95% CI)
18–54	13.0	4.0	9.0 (6.0–12.8)
55–64	22.3	10.1	12.2 (10.3–14.3)
65–74	36.7	23.3	13.3 (9.0–18.6)
≥75	85.0	57.8	27.2 (13.5–45.5)



Age, years	Mean mortality rate/1000 person-years: ACR, mg/g 10	300	Difference (95% CI)
18–54	4.0	11.6	7.5 (4.3–11.9)
55–64	10.3	22.5	12.2 (7.9–17.6)
65–74	22.0	44.7	22.7 (15.3–31.6)
≥75	49.6	84.0	34.3 (19.5–52.4)

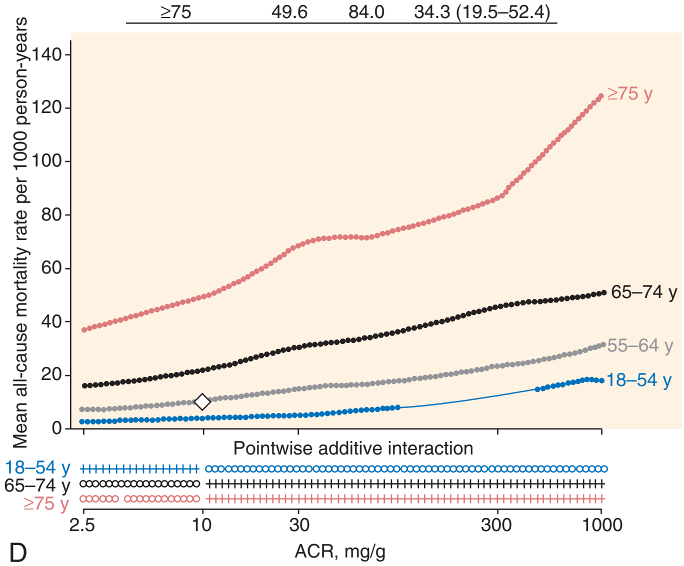


Fig. 19.10 Adjusted hazard ratios *(HRs)* for all-cause mortality and mean mortality rates according to estimated glomerular filtration rate *(eGFR)* and albumin-creatinine ratio *(ACR)* within each age category. *Solid circles* denote statistical significance ($P < .05$) compared with the reference *(diamond)* eGFR of 80 mL/min/1.73 m^2, or ACR of 10 mg/g within each age category in (A) and (B) and compared with the age category of 55 to 64 years in (C) and (D). *Plus signs* and *open circles* at the bottom of each graph represent significantly positive (greater effect size) and negative (smaller effect size) pointwise interactions ($P < .05$), respectively, compared with age 55 to 64 years. Gaps indicate no significant pointwise interaction. Models are meta-analyses of general population and high-risk cohorts adjusted for gender, race, body mass index, systolic blood pressure, total cholesterol, history of cardiovascular disease, diabetes, smoking status, and albuminuria (A and C) or eGFR (B and D). (From Hallan SI, Matsushita K, Sang Y, et al. Age and association of kidney measures with mortality and end-stage renal disease. *JAMA.* 2012;308:2349–2360.)

cell trait, present in 7% of African Americans but virtually absent in whites, also confers up to a twofold higher risk of ESKD, although less reliably across studies.[14,15] Both *APOL1* high-risk status and sickle cell trait are present in greater proportions in the ESKD population than in the general population.[87,88]

Other discovered genetic variants may also have effects on the development and progression of CKD, but they have generally had much smaller population prevalences or effect sizes than *APOL1* risk alleles and the sickle cell trait.[89–92] For example, genetic variation in *UMOD*, the gene coding for uromodulin, expressed in the thick ascending limb of the loop of Henle, may play a role in CKD. Uromodulin is the most commonly found protein in normal urine and likely affects renal salt handling.[93,94] Common variants in the promoter region of *UMOD* have been associated with increased risk of CKD, CKD progression, ESKD, and hypertension in genome-wide association studies, albeit with a difference in

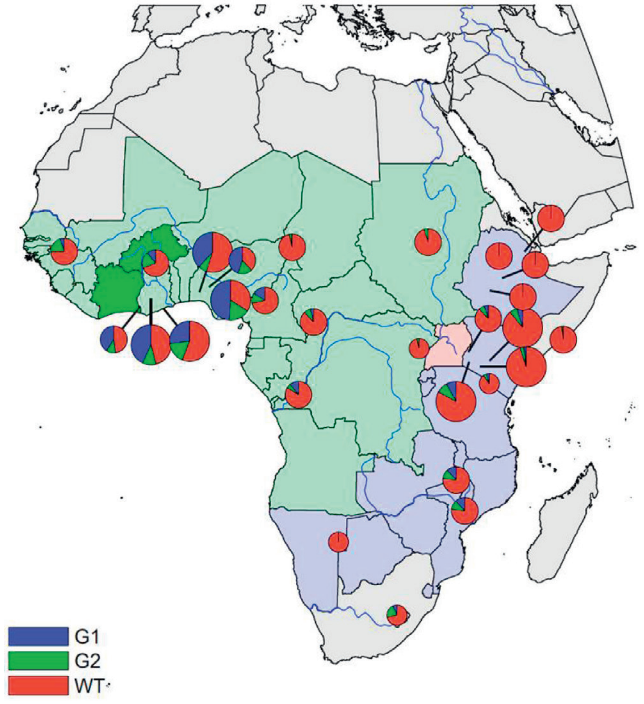


Fig. 19.11 Distribution of the G1 and G2 *APOL1* variants across Africa. Allele frequencies of the G1 and G2 variants are indicated as *blue* and *green wedges*, respectively. Circle size reflects the number of individuals genotyped: small, <10 individuals/20 chromosomes; medium, 10–100 individuals/20–200 chromosomes; large, >100 individuals/200 chromosomes. Countries are shaded according to the subspecies of *Trypanosoma brucei* that cause African sleeping sickness. *Darker green,* gambiense types 1 and 2; *light green,* gambiense type 1; *pink,* both rhodesiense and gambiense type 1; *purple,* rhodesiense. (From Thomson R, Genovese G, Canon C, et al. Evolution of the primate trypanolytic factor APOL1. *Proc Natl Acad Sci U S A.* 2014;111:E2130–E2139.)

the rate of eGFR decline in the range of 0.15 mL/min/1.73 m^2 per year.[91,95–101] Genome-wide association studies have also identified genomic regions associated with more specific disease types, such as IgA nephropathy and idiopathic membranous nephropathy.[20,102] Many genetic variants underlying monogenic disease have also been determined, such as variants in *PKD1* or *PKD2* causing autosomal dominant polycystic kidney disease and more than 20 genes involved in syndromic and nonsyndromic CAKUT (a leading cause of kidney disease in children), but most of these genotypes are rare (<1 in 1000 for dominant diseases and <1 in 40,000 for recessive diseases).[22,103,104]

Among clinical risk factors, the presence of diabetes mellitus and hypertension appear to be the most robustly associated with CKD. The presence of diabetes mellitus was a risk factor for the development of CKD after full or partial nephrectomy[105] and also in several community-based studies.[106,107] Diabetes has also been linked to the development of kidney failure,[108] and diabetes is considered the most common cause of ESKD in the world today.[24] Evidence that tight glycemic control helps forestall the development of mild to moderate albuminuria in clinical trials further advances the notion that the presence of diabetes mellitus is on the causal pathway for the development and progression of many types of CKD.[109] Interestingly, among people with CKD, diabetes is a weaker risk factor for ESKD once the GFR and albuminuria are accounted for.[64]

In contrast to the well-accepted role of diabetes in the development of CKD, there is ongoing debate as to whether hypertension is a cause or merely a consequence of CKD. Many observational studies have linked higher blood pressure to the development and progression of CKD,[39,108,110] but evidence from randomized controlled trials that lowering blood pressure helps forestall CKD progression is inconsistent.[111–113] The MDRD study ($N = 840$) showed no benefit of the lower blood pressure goal on GFR decline during the trial period (mean, 2.2 years) but a beneficial effect on ESKD during longer follow-up. The African American Study of Kidney Disease and Hypertension ($N = 1094$) showed no benefit of the lower blood pressure goal on GFR decline over a mean of 4 years or on ESKD during longer follow-up.[111] However, both these studies suggested a larger benefit of the lower blood pressure goal in patients with higher proteinuria.[114,115] The REIN-2 trial did not show a difference in progression to ESKD with a lower blood pressure goal, but this study achieved the lower blood pressure goal using a calcium channel blocker, which increases proteinuria, which may have blunted any beneficial effect.[99] In PKD, the HALT-PKD study showed a beneficial effect of a lower blood pressure goal on enlargement in total kidney volume but not on eGFR decline.[116] In SPRINT, a lower blood pressure goal appeared to result in a lower eGFR over the first 18 months, but eGFR trajectories remained relatively stable after that time, with a large beneficial effect of lower blood pressure on death and CVD, despite the higher risk of incident CKD.[117]

Another increasingly prevalent risk factor for CKD is obesity.[32] In a study of 320,252 participants in a U.S. health plan, obesity was associated with more than a threefold risk of developing ESKD over a mean of 26 years of follow-up. Contemporary studies with shorter follow-up have shown slightly more modest risks of CKD stages G4–G5 associated with obesity, such as one in the United Kingdom, which estimated a twofold risk associated with body mass index (BMI) between 30 and 35 kg/m^2.[118] Similarly, a meta-analysis of ESKD risk among over 4 million potential kidney donor candidates showed a 1.16 times higher risk per 5 kg/m^2 higher BMI over 30 kg/m^2.[119] The risk of CKD progression associated with obesity may be smaller still; in one cohort of Swedish people with CKD stages G4–G5, BMI was unrelated to subsequent ESKD. On the other hand, weight loss achieved by lifestyle modification or bariatric surgery has been associated with a reduction in albuminuria and preservation of eGFR.[120–122] For example, in Look AHEAD, a clinical trial in 5145 participants with diabetes, an intensive lifestyle intervention resulted in 4 kg of weight loss compared with the control arm and a 31% reduction in the development of CKD in very high-risk categories, as defined by KDIGO. Obesity may contribute to CKD as a risk factor mediated by the development of diabetes and hypertension or with independent causal effects. In animal models, obesity results in glomerular hypertension and hyperfiltration, a proinflammatory milieu, and reduced adiponectin levels, all of which may contribute to CKD risk.[123]

There have been localized epidemics of kidney disease, such as Balkan nephropathy, those due to Chinese herbal supplements (now attributed to aristolochic acid[124]), and the more recent Mesoamerican nephropathy,[125] which point to environmental factors that may additionally alter the risk of ESKD in populations.

END-STAGE KIDNEY DISEASE

EPIDEMIOLOGY

In the United States, data regarding ESKD are generally excellent due to the establishment of comprehensive and well-maintained kidney disease registries by the National Institutes of Health in 1989.[126] Today, the Annual Data Reports of the U.S. Renal Data System (USRDS) and the Scientific Registry of Transplant Recipients (SRTR) provide extensive epidemiologic information on incident and prevalent ESKD based on required reporting by providers of dialysis and transplantation services. As such, the epidemiology of ESKD reflects the clinical practice of initiation of KRT as well as the progression of CKD to kidney failure. Information on untreated kidney failure—GFR < 15 mL/min/1.73 m^2 not treated by KRT—is not captured by the USRDS or SRTR. Some data sources have suggested that untreated kidney failure could represent a substantial number of kidney failure cases, particularly among the very old.[47,66] Causes of kidney disease among those with ESKD are ascertained from the Centers for Medicare & Medicaid Services Form 2728, but are frequently missing and have not been validated.[127] Because dialysis is the predominant mode of KRT, the epidemiology of ESKD largely reflects the epidemiology of dialysis. Specific topics related to the epidemiology of kidney transplantation are covered in a separate section.

PREVALENCE

As of December 31, 2014, there were 678,383 prevalent cases of ESKD in the United States, or a rate of 2067/million population.[24] Both figures increased over the preceding year (3.5% and 2.6%, respectively). For reference, in 1996, the number of cases of ESKD was 303,311, or 1095 cases/million population (Fig. 19.12). Much of the growth in prevalence was attributable to the aging of the population (ESKD is more common in older populations) and growth in ESKD in those aged 75 years and older, in whom the sex- and race-adjusted prevalence of ESKD increased from 2989/million population in 1996 to 6243/million population in 2014. The growth in prevalence is also partly due to a decline in mortality, which decreased from 186/1000 patient-years to 137 deaths/1000 patient-years over the corresponding time period.

In the United States, approximately 70% of prevalent ESKD cases were treated by dialysis, either hemodialysis (63.1%) or peritoneal dialysis (6.9%). Most ESKD cases were attributed to diabetes or hypertension (37.8% and 25.3% of the age-, gender-, and race-adjusted prevalence, respectively), followed by glomerulonephritis (16.4%) and cystic kidney disease (3.8%).

In 2014, the USRDS collected ESKD data from 60 different countries, enabling comparisons of both the absolute counts of ESKD cases as well as the prevalence per million population. In total, there were 2,217,350 patients with ESKD counted globally. Patients in the United States accounted for 30% of the total, followed by Japan (14%) and Brazil (7%). The prevalence of ESKD within the population varied substantially across the participating country, ranging from 113/million population in Bangladesh to 3219 cases/million population in Taiwan. Japan had the second highest prevalence, 2505/million population. All countries providing annual data since 2001 (N= 32) showed increases in prevalence (median, 48%). The increase was greatest in the Philippines and Thailand, where the percentage changes from 2001 to 2014 were 1092% and 902%, respectively.

Large variations also existed among countries with respect to the distribution of KRT modality (Fig. 19.13). In Norway, 72% of prevalent ESKD patients are treated with a functioning

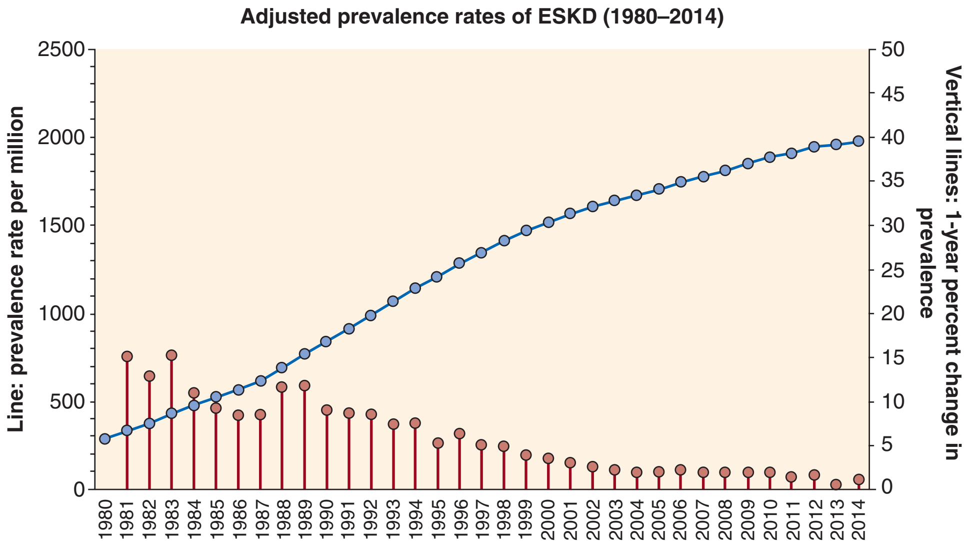


Fig. 19.12 Trends in adjusted* ESKD prevalence (per million; trend lines; scale on left), and annual percentage (%) change in adjusted* prevalence of ESKD (*vertical lines;* scale on right) in the U.S. population, 1980–2014, adjusted for age, gender, race, and ethnicity. The standard population was the U.S. population in 2011. (Data from U.S. Renal Data System. 2014 USRDS annual data report: epidemiology of kidney disease in the United States. https://www.ajkd.org/article/S0272-6386(15)00744-1/fulltext; and U.S. Renal Data System. 2016 USRDS annual data report: epidemiology of kidney disease in the United States. https://www.ajkd.org/article/S0272-6386(16)30703-X/fulltext.)

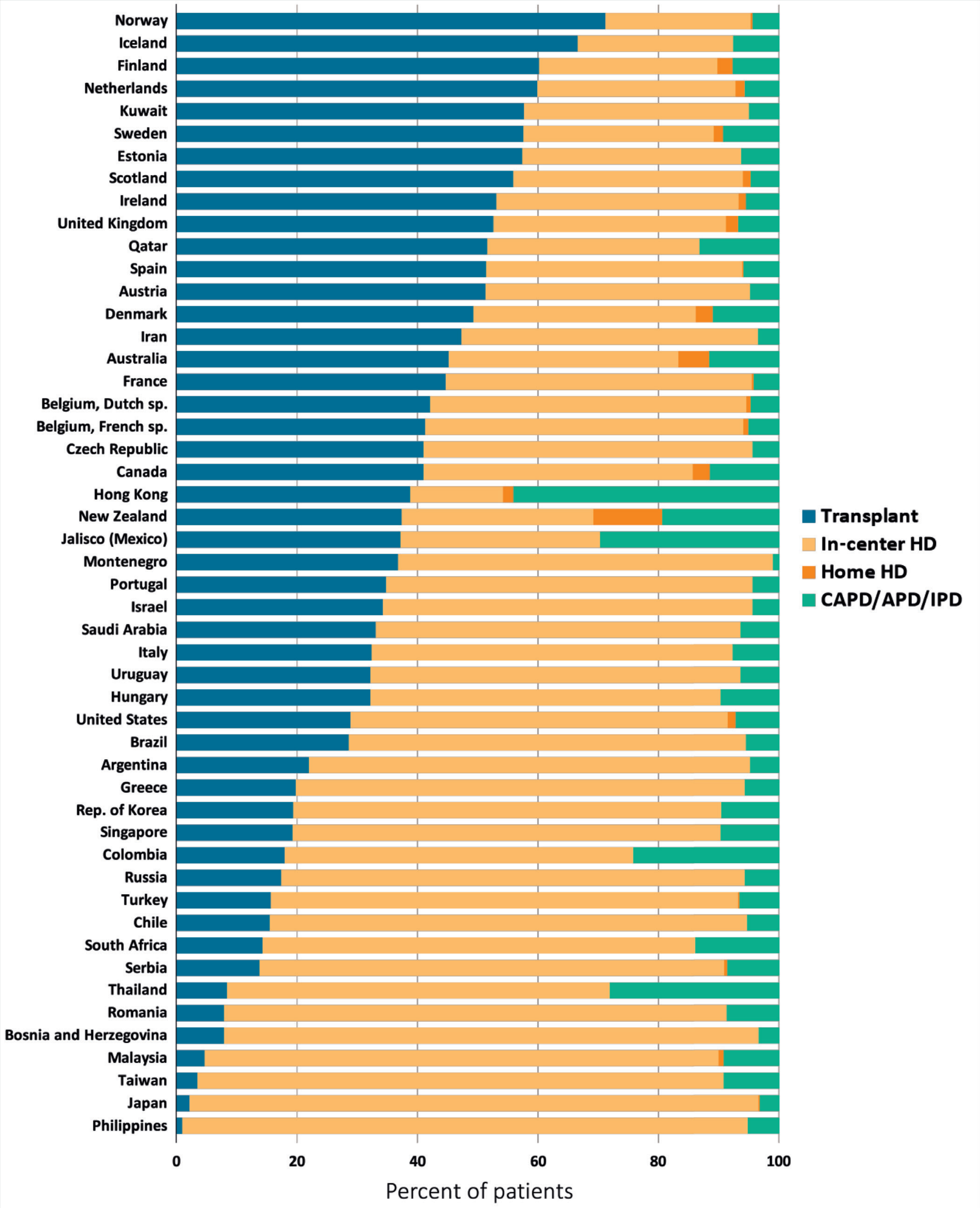


Fig. 19.13 Percentage distribution of type of kidney replacement therapy modality used by patients with end-stage kidney disease. By country, in 2014. (From U.S. Renal Data System. 2016 USRDS annual data report: epidemiology of kidney disease in the United States. https://www.ajkd.org/article/S0272-6386(16)30703-X/fulltext.)

transplant, 24% receive in-center hemodialysis, and 4% receive peritoneal dialysis. Peritoneal dialysis is most common in Mexico, where 44% receive peritoneal dialysis, 39% have a functioning transplant, and 17% receive in-center or home hemodialysis. In contrast, in-center hemodialysis consitutes 94% of ESKD in both Japan and the Philippines. Additional information on differences in hemodialysis practice by country was collected by the Dialysis Outcomes and Practice Patterns Study (DOPPS) between June 1996 and March 2012.[128] Interestingly, in the population of 35,964 hemodialysis patients sampled from 12 countries, the proportion of women was much lower than that of the general population. This appeared attributable to a greater incidence of ESKD among men rather than better survival, because the male-to-female mortality ratio was close to 1.

There is evidence to suggest that the variation in ESKD prevalence has much to do with the resources available to provide KRT. A recent study has suggested that the gap between people worldwide who require KRT and those who undergo KRT was at least 2.3 million in 2010.[69] The authors estimated the prevalence of ESKD undergoing KRT to be 1839 cases/million people in North America, followed by 719 cases/million people in Europe, 686 cases/million in Oceania, and 626 cases/million in Latin America. Asia, which was reported to have an ESKD prevalence of 232 cases/million population, was estimated to have the highest absolute number of people with ESKD, at nearly 1 million cases in 2010. Gross national product was highly associated with prevalence, with the highest-income countries reporting a much higher prevalence of patients undergoing KRT. Interestingly, this relationship was much stronger than the relationship between the prevalence of diabetes and the prevalence of patients undergoing KRT (Fig. 19.14).

INCIDENCE

The incidence, or rate of new cases, of ESKD is also reported annually by the USRDS.[24] In the United States, the number of incident cases of ESKD in 2014 was 120,688, or an incidence rate of 370/million per year. These figures are 2.2% and 1.1% increases, respectively, over the prior year. In 1996, the number of incident cases of ESKD was 77,018, or 278 cases/million population. Much of this increase in incidence rates can be attributed to the aging of the U.S. population and the increase in incidence of ESKD among those older than 75 years, for whom the race- and gender-adjusted incidence rate increased from 1203 cases/million population in 1996 to 1556 cases/million population in 2014.

Most of the new ESKD cases initiated KRT using hemodialysis (87.9%), whereas 9.3% used peritoneal dialysis and 2.6% received a kidney transplant without prior dialysis. These proportions have remained relatively stable over the past decade. The adjusted incidences of ESKD due to diabetes, hypertension, and cystic disease have remained relatively consistent with respect to each other; the adjusted incidence of glomerulonephritis declined slightly. The adjusted incidence rate ratios for blacks versus whites in 2014 was 3.1; for Native Americans and Asians, the ratio versus whites was 1.2.

When extended to estimate the lifetime incidence of ESKD, racial and ethnic disparities remain.[43,129,130] One study using 2013 USRDS data has suggested that the lifetime incidence of ESKD is 3.1% for a non-Hispanic white man, 6.2% for a Hispanic man, and 8.0% for a non-Hispanic black man.[129] Furthermore, these disparities changed very little over the previous decade.

When compared with other countries, the incidence of ESKD in the United States was among the highest, trailing only Taiwan and the Jalisco region of Mexico in 2014 (Fig. 19.15).[24] Other high-ESKD incidence countries included Thailand, Singapore, Japan, Korea, and Malaysia. Among the countries represented, the lowest incidence rate/million population per year was in Bangladesh, with 49 ESKD cases/million per year; Iceland, with 58 cases/million per year; and Russia, with 60 cases/million per year. Trends in the incidence of ESKD varied by country, with many of the Northern European countries experiencing a lower ESKD incidence over time. In contrast, there were large increases in incidence rates in many of the developing countries (Fig. 19.16).

Interestingly, the proportion of new cases attributed to diabetes also varied by country. In Asia, Latin America, and the United States, most countries reported over 40% of the new cases as attributable to diabetes. In contrast, many of the European countries reported only 20% of the new cases as being attributed to diabetes (Fig. 19.17). In addition, in all countries, there was a disparity in new ESKD by gender. Men had higher rates of new ESKD than women, sometimes even twofold higher.

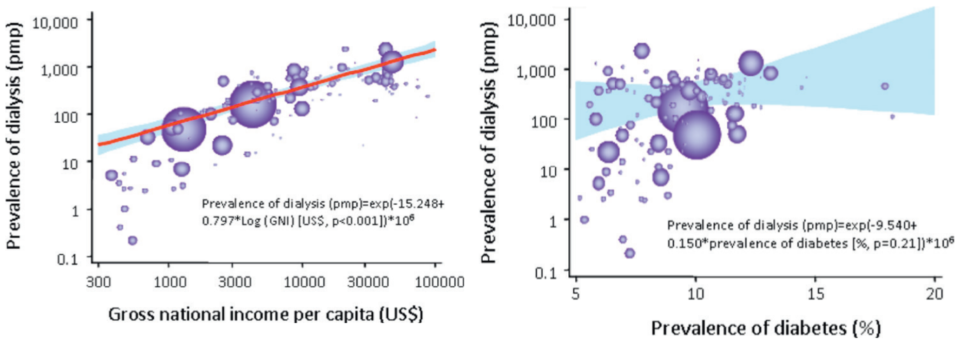


Fig. 19.14 Association between prevalence of end-stage kidney disease (ESKD) treated by dialysis in 123 countries and (gross national income per capita [left], prevalence of diabetes [right]). In both graphs, China and India are represented by the largest and second largest circles, respectively. (From Liyanage T, Ninomiya V, Jha V, et al. Worldwide access to treatment for end-stage kidney disease: a systematic review. *Lancet.* 2015;385[9981]:1975–1982.)

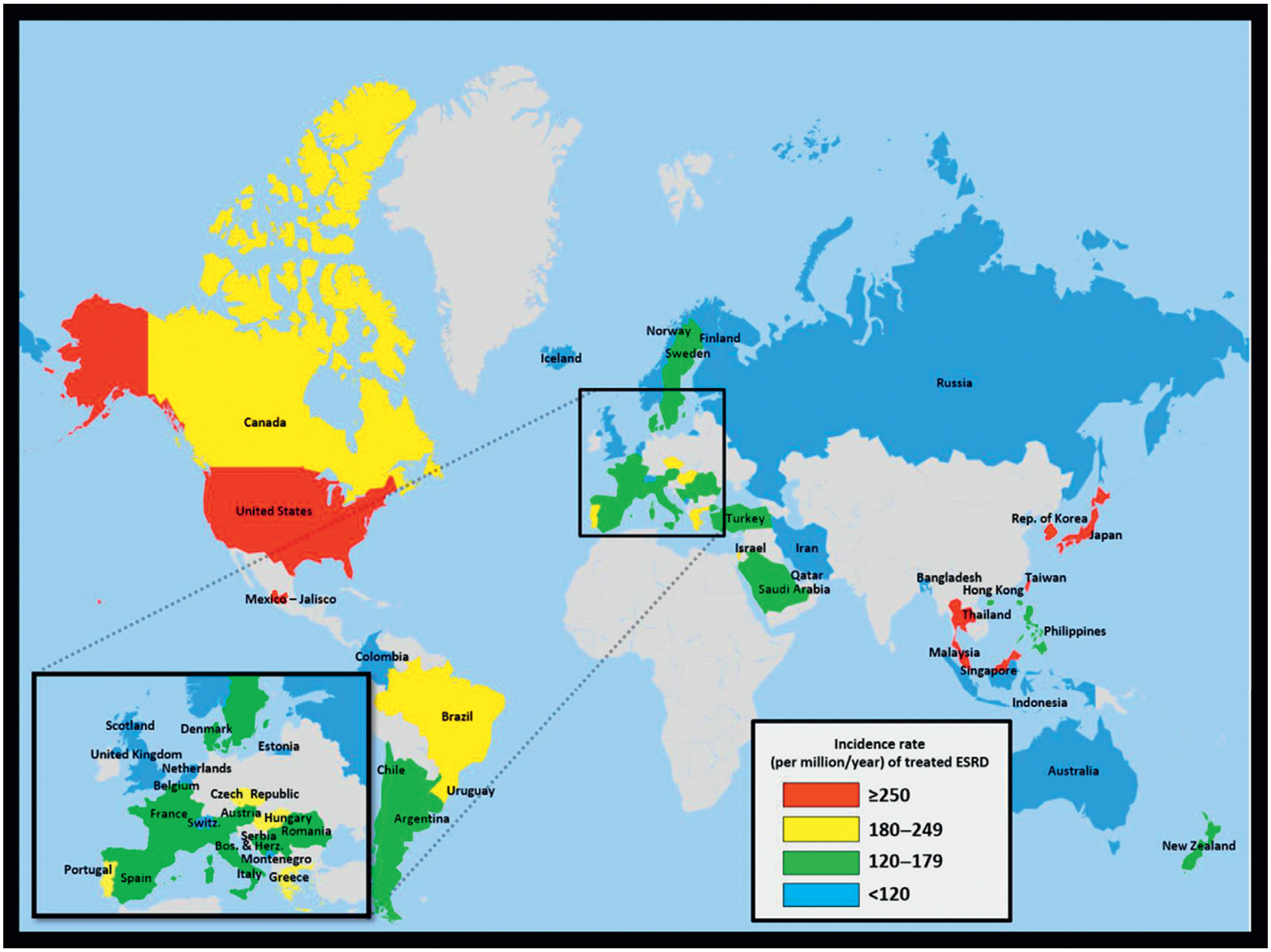


Fig. 19.15 Geographic variations in the incidence rate of treated end-stage kidney disease (*ESKD;* per million population/year), by country, 2014. (From U.S. Renal Data System. 2016 USRDS annual data report: epidemiology of kidney disease in the United States. https://www.ajkd.org/article/S0272-6386(16)30703-X/fulltext.)

OUTCOMES

Age-adjusted mortality in ESKD is much higher than in the general population.[24,131,132] In 2014, the USRDS reported that the adjusted mortality rate for ESKD was 136 deaths/1000 patient years. For patients receiving hemodialysis, adjusted mortality rates were 169 deaths/1000 patient-years; for peritoneal dialysis, the rate was 157 deaths/1000 patient-years. In the year after dialysis initiation, the risk of death peaked at 2 months for patients receiving hemodialysis and increased slowly over the year for patients receiving peritoneal dialysis. Approximately 41% of all deaths among dialysis patients in 2012 were attributed to CVD.

Mortality rates in incident dialysis patients vary by age and underlying comorbid conditions. Early mortality is more common among people with lower serum albumin and phosphorus levels, as well as those with congestive heart failure and cancer.[133] Although racial differences in dialysis mortality have been noted, particularly in older age groups, they may be due in part to differences in the rate of transplantation, the level of residual kidney function at dialysis initiation, and the presence and severity of comorbid conditions.[134]

Mortality rates in incident dialysis patients also vary by country. In patients selected for the DOPPS, the mortality rate ranged from 17 deaths/100 patient-years in the first 120 days in Japan to 33.5 deaths/100 patient-years in the first 120 days in Belgium.[131] Long-term mortality was even more disparate: after the first year on dialysis, there were 5.2 deaths/100 patient-years in Japan compared with 19.9 deaths/100 patient-years in Belgium. It should be noted that the proportion of patients 75 years of age and older was smaller in Japan; however, adjusted mortality rates were also lower in Japan than in any other included countries. In general, European and North American long-term mortality rates were fairly similar, with the United States reporting the highest adjusted mortality rates (Fig. 19.18).

It should also be noted that mortality on dialysis is sometimes related to withdrawal of therapy. In Australia and New Zealand, nearly 40% of deaths within the first 120 days were coded as being due to withdrawal from care, compared

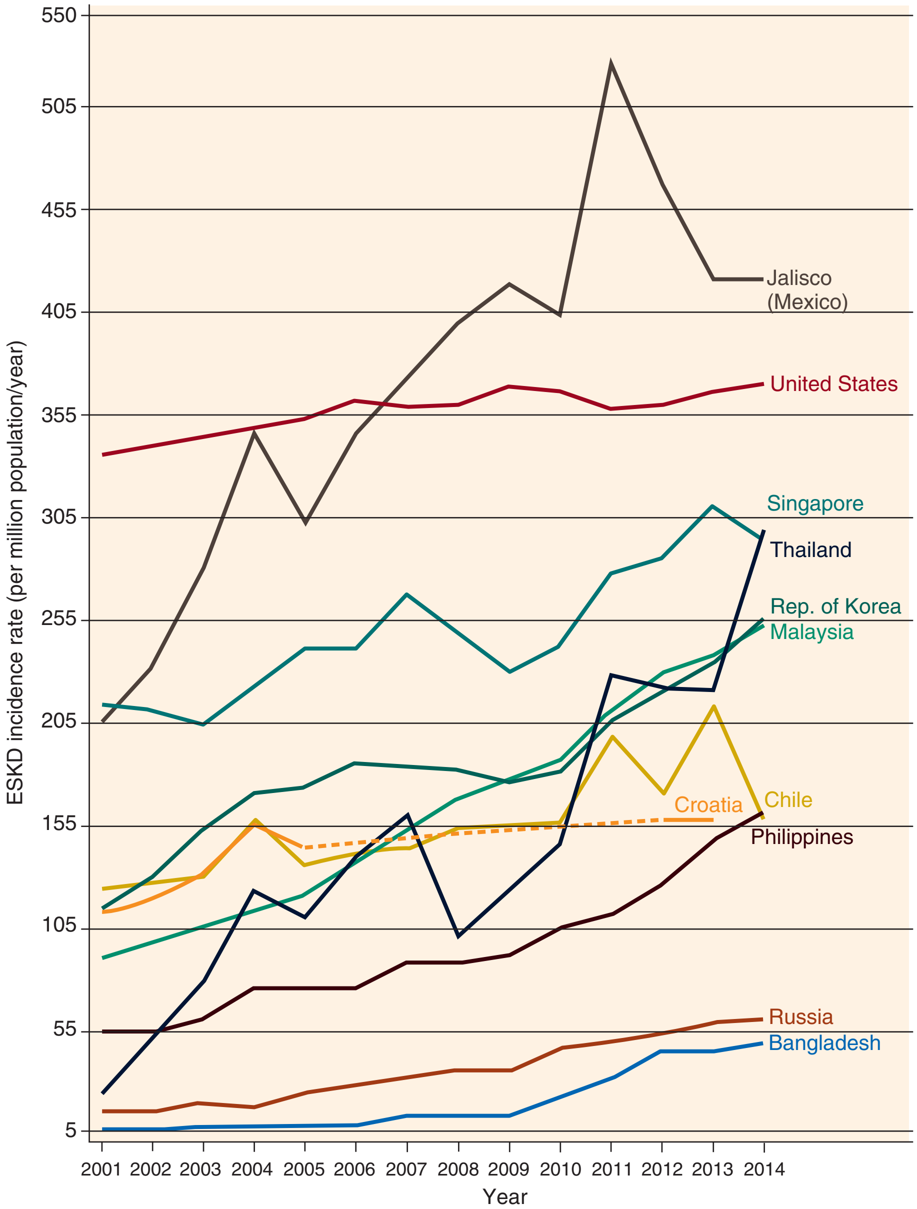


Fig. 19.16 Trends in the incidence rate of treated end-stage kidney disease (*ESKD;* per million population/year), by country, 2001– 2014. The 10 countries with the highest percentage rise in ESKD incidence rate in 2001–2002 versus that in 2013–2014, plus the United States. (From U.S. Renal Data System. 2016 USRDS Annual data report: epidemiology of kidney disease in the United States. https://www.ajkd.org/article/S0272-6386(16)30703-X/fulltext.)

with much lower rates in regions such as Italy, France, and Germany.[131]

Trends in mortality have suggested that dialysis patients in the United States are living longer compared with previous decades, with the overall mortality rate decreasing by 32% since 1996. Although there have been numerous improvements in dialysis treatment, their relative contribution to the trend toward lower mortality is uncertain. With the exception of antihypertensive treatment, controlled trials evaluating single interventions have not shown an effect on all-cause mortality in dialysis patients.[135–143] In our view, this may be due in part to the difficulty of showing a reduction in all-cause mortality with interventions targeting a single risk factor. The adoption of multiple interventions in practice may achieve this goal.

RISK FACTORS

Risk factors for the development of ESKD demonstrate some notable differences from those that portend a higher risk of incidence of earlier stages of CKD. Most notably, age has often demonstrated an inverse association with ESKD risk, adjusted for other factors.[65,144,145] Given the strong association between older age and incidence of CKD, the reverse association is somewhat surprising. Possible explanations are that the competing event of mortality becomes much more common in

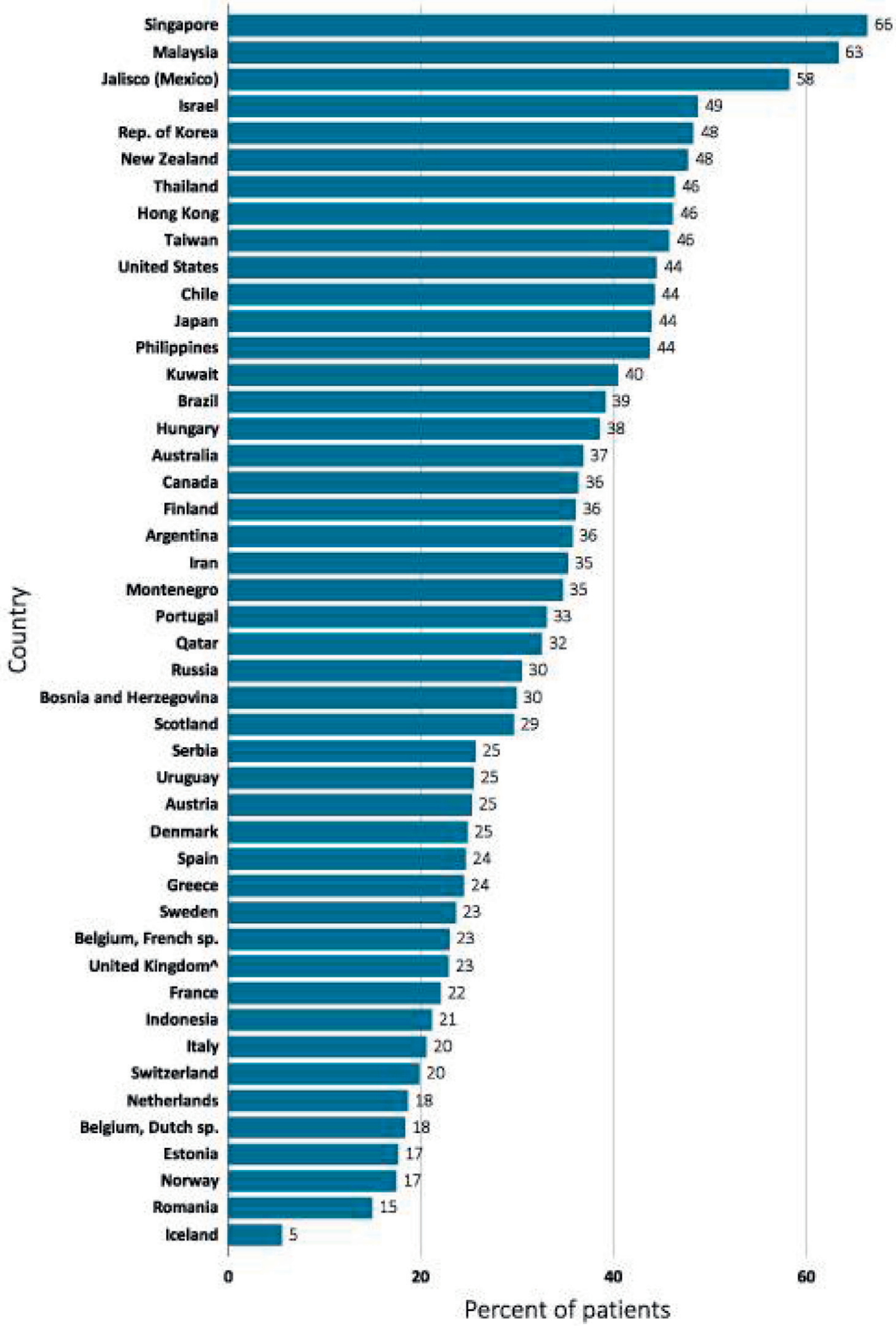


Fig. 19.17 Proportions of incident end-stage kidney disease (ESKD) due to diabetes, by country, 2014. (From U.S. Renal Data System. 2016 USRDS annual data report: epidemiology of kidney disease in the United States. https://www.ajkd.org/article/S0272-6386(16)30703-X/fulltext.)

older adults, and many older adults may choose not to initiate dialysis. Indeed, in Canada, people who were 85 years or older with an eGFR from 15 to 29 mL/min/1.73 m^2 experienced 20 cases of untreated kidney failure/1000 patient-years and 1.5 cases/1000 patient-years of treated kidney failure (i.e., undergoing dialysis or transplantation).[66] In contrast, for people ages 18 to 44 years with a similar eGFR, the rates were 3.5 cases/1000 person-years and 17 cases/1000 person-years of untreated and treated kidney failure, respectively. Similar observations have been demonstrated in the United States.[47] Finally, it is possible that eGFR decline is slower among older compared with younger individuals.[146]

Gender and race are additional risk factors that display somewhat disparate associations with the incidence of earlier stages of CKD compared with ESKD. For example, although women may be at a slightly higher risk for eGFR < 60 mL/min/1.73 m^2,[147] men make up a greater proportion of incident and prevalent ESKD patients.[24] The higher risk of ESKD among men persists, even after adjusting for demographic and clinical factors. Reasons for ESKD differences by gender may

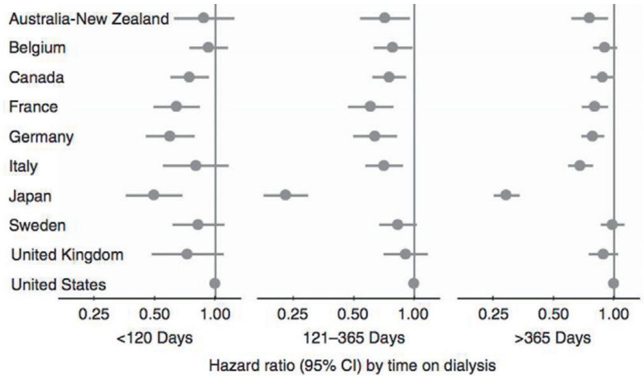


Fig. 19.18 Mortality in each DOPPS (Dialysis Outcomes and Practice Patterns Study) country versus the United States by time on hemodialysis (HD). Model adjusted for age, gender, race, and diabetes as the cause of end-stage kidney disease, stratified by study phase (N = 86,886 HD patients from DOPPS census [2002–2008]). *CI,* Confidence interval. (From Robinson BM, Zhang J, Morgenstern H, et al. Worldwide, mortality risk is high soon after initiation of hemodialysis. *Kidney Int.* 2014;85:158–165.)

stem from hormonal differences[148] or simply from differences in care delivery and preference. For example, women tend to start dialysis at a lower eGFR than men.[149] In contrast, despite later initiation of dialysis among African Americans, African Americans have nearly a fourfold higher ESKD risk than Caucasians.[24,43,149] These disparities persist after adjustment for socioeconomic status, clinical characteristics, and access to health care.[77,150,151] The age- and gender-adjusted incidence of ESKD among African Americans was 876.7 cases/million per year in 2014, compared with 357.3 cases/million among Asians, 331.1 cases/million among Native Americans, and 286.3/million among U.S. whites.

As discussed previously, diabetes mellitus is considered to be the major cause of ESKD worldwide, in part due its growing incidence and prevalence.[24,152] In Singapore, 66% of incident ESKD cases in 2014 were attributed to diabetes; the corresponding figure in the United States was 46%. In a USRDS survey of 46 world regions, only 6 reported that less than 20% of ESKD cases were due to diabetes (the Netherlands, Dutch-speaking Belgium, Estonia, Norway, Romania, and Iceland).[24] Furthermore, diabetes mellitus can be considered as a risk factor for both CKD incidence and disease progression. Diabetic CKD is frequently characterized by the presence of albuminuria, which is also a risk factor for CKD progression to ESKD.[63,64] The second most common listed cause for ESKD in the United States is hypertension, although it is believed that many of the cases ascribed to hypertension may be due to other factors, such as *APOL1* risk alleles or parenchymal disease.[153] Obesity has also variably been associated with increased ESKD risk, with older studies with longer follow-up demonstrating stronger associations than more contemporary studies.[119,154]

Geographic region accounts for some of the differences in ESKD risk as well. In a validation of a kidney failure risk equation calculator in 31 international cohorts, only the region of cohort was observed to improve calibration for a model that included age, gender, eGFR, and albuminuria.[63] Therefore, a regional calibration factor was introduced that incorporated a lower baseline risk by 32.9% at 2 years and 16.5% at 5 years for countries outside North America (see kidneyfailurerisk.com). The use of a calibration factor for geographical region is consistent with a recent systematic review, which reported differences in ESKD prevalence by region. In that study North America had the highest prevalence, at 1839 cases/million people, followed by Europe, at 719 cases/million people.[69] These differences appeared driven in large part by income of the country, with an estimated 74-fold difference in prevalence between high-income and low-income countries.

KIDNEY TRANSPLANTATION

PREVALENCE

In the United States, approximately 30% of prevalent ESKD cases have a functioning kidney transplant, corresponding to 200,907 people in 2014.[24] The number of prevalent cases has more than doubled over the past 2 decades, from 83,199 in 1996. The prevalence is currently one of the highest in the world, at 630 people with a functioning kidney transplant/million population in the United States, but other countries such as Norway (657 cases/million) and Portugal (642 cases/million) also have a high prevalence. In general, the prevalence of those with functioning kidney transplants is highest in North America and Europe and lowest in South America and parts of Asia. Prevalence in all assessed countries has increased over the past decade. As a proportion of prevalent ESKD cases, patients living with a functioning transplant was fairly stable in most countries. The highest proportions were generally found in Northern Europe, where more than 50% of ESKD patients were living with a functional kidney transplant.

INCIDENCE

In the United States, there are approximately 18,000 transplantations/year, a number that has remained relatively stable over the past decade, despite a growing prevalence of patients with ESKD (Fig. 19.19).[24] In 2014, the proportion of transplanted organs that originated from a deceased donor was 69%, a figure that has increased slightly over the years, with a concomitant decline in living donor transplants.[155] Approximately 30% of living donor transplant procedures were performed prior to the initiation of dialysis.[156] The median waiting time for transplantation was 3.4 years in 2009 for both living and deceased donor transplants combined. Waiting time was significantly different by blood type, panel-reactive antibodies (PRA), and region. Using 2014 data, kidney transplantation rates/1000 dialysis patient-years also varied—highest among people aged 21 years and younger (321) and lowest among those aged 75 years and older (4). On average, women had slightly lower transplantation rates than men (32 vs. 39/1000 dialysis patient-years) as did black patients compared with other races (25 vs. 41, 27, and 42 for white, Native American, and Asian dialysis patients, respectively).

Internationally, there is even greater variation in kidney transplantation rates, due in part to differences in health care systems, availability of organs, and cultural beliefs. In 2014, the highest transplantation rates/1000 dialysis patients were seen in Northern Europe, with Norway and the Netherlands the highest, at 205 and 154 transplants/1000 dialysis

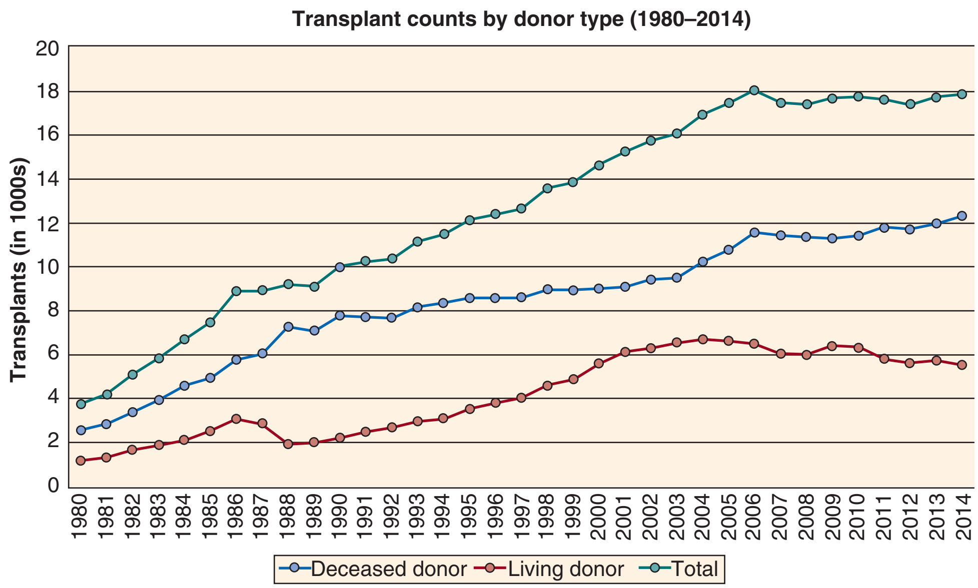


Fig. 19.19 Number of kidney transplants by donor type, 1980-2014. (Data from U.S. Renal Data System. 2013 Annual data report: atlas of chronic kidney disease and end-stage renal disease in the United States. https://www.ajkd.org/article/S0272-6386(13)01544-8/pdf; and U.S. Renal Data System. 2016 USRDS annual data report: epidemiology of kidney disease in the United States. https://www.ajkd.org/article/S0272-6386(16)30703-X/fulltext.)

patients, respectively. In Norway, nearly all patients younger than 65 years and most patients older than 65 years received transplants within 4 years of initiating KRT.[157]

OUTCOMES

Randomized controlled trials have not been performed, but kidney transplantation is generally accepted as the preferred KRT modality, with lower mortality and better quality of life compared with hemodialysis or peritoneal dialysis.[158] These benefits are thought to extend to recipients of both living and deceased donor organs, including those who receive organs with so-called marginal characteristics.[159–161] Benefits of transplantation are also seen in older recipients; for example, patients older than 70 years who received transplants had a 41% lower risk of death compared with wait-listed counterparts in adjusted analysis.[162] Whereas cardiovascular mortality increases with dialysis vintage, there is no such increase in cardiovascular events with kidney transplantation vintage, suggesting that some of the mortality benefit seen with transplantation may be a reduction in the development of CVD.[163]

On the other hand, kidney transplantation is not without risk. The leading cause of death is CVD. Among risk factors for CVD, new-onset diabetes mellitus after transplantation is of particular relevance because it may be increased by the calcineurin inhibitor tacrolimus, a commonly used agent for immunosuppression. New-onset diabetes is variably captured but has been reported in as many as 50% of transplant recipients, with a higher risk among older recipients.[164] Infections are the second leading cause of death in transplant recipients, and the infection rate in the first 3 years after transplantation is estimated at 45 infections/100 recipient-years of follow-up.[165,166] Cancer is the third leading cause of death, with the incidence of cancer mortality two- to threefold higher than the general population.[167,168]

DISPARITIES IN KIDNEY TRANSPLANTATION

Several international studies have suggested disparities in rates of kidney transplantation by socioeconomic status and race, particularly for living donor kidney transplants.[169,170] In the United States, disparities have been documented in transplant referral, recipient candidate approval, placement on the waiting list, time on the waiting list, and allograft survival.[170–175] It has been estimated that eligible African American transplant candidates receive 35% fewer transplants than eligible Caucasian transplant candidates.[176] A study in Australia has found that recipients from socioeconomically advantaged areas were more likely to receive a living kidney donor transplant than those of lower socioeconomic status, although rates of deceased donor transplantation were similar.[177] Some of the disparities in living donor transplantation may be due to increased risk of CKD in family members due to the presence of genetic risk variants. Similarly, the presence of the *APOL1* high-risk genotype in the allograft (but, interestingly, not the recipient), is a risk factor for shortened allograft survival.[178]

ACUTE KIDNEY INJURY

EPIDEMIOLOGY

There are many similarities in the epidemiology of AKI and CKD, in part because the definitions for both include a criterion based on a measure of GFR (see Table 19.1). The urine output criterion is unique to the definition of AKI, but it is also relatively recent, not captured in much of the previous literature, and has been criticized for both logistic and theoretic reasons.[179,180] Studies of AKI typically capture disease by changes in creatinine level (with a wide range of definitions) or by relying on billing codes, such as those classified in the *International Classification of Disease, Ninth Clinical Modification.*[179] Studies using diagnostic codes to identify AKI events are particularly problematic, because labeling an event as AKI depends on the provider's perception of what constitutes AKI. Using a creatinine-based definition of AKI, the sensitivity of diagnostic codes has been shown to vary by era and patient age.[181]

INCIDENCE

By definition, AKI is a transient disease; kidney disease lasting longer than 3 months is defined as CKD.[3,6] Thus, AKI can be described only through incidence rather than prevalence. One study of community-dwelling people in Northern California has estimated the incidence rate to be 522.4 cases of AKI/100,000 person-years and 29.5 cases of dialysis-requiring AKI/100,000 person-years.[182] A population-based study from Scotland has estimated the AKI and AKI requiring dialysis rates as 214.7 and 18.3 cases/100,000 patient-years, respectively.[183] In a Canadian cohort of patients with both creatinine and urine protein measurements, incidence rates were much higher, ranging from 100 to 11,700 cases of hospitalized AKI/100,000 person-years, depending on the level of GFR and proteinuria.[184] In patients with Medicare insurance and older than 65 years, 4.0% had a hospitalization with AKI over the course of 1 year, as identified by a billing code for AKI.[24]

In a hospitalized population, the rate of AKI may be even higher. A meta-analysis of hospitalized patients worldwide has estimated that almost 25% had AKI ascertained using the KDIGO definition; approximately 10% of cases required KRT.[185] A similar estimate was derived from U.S. Department of Veterans Affairs patients and, notably, only 49% of those with AKI ascertained by the serum creatinine criterion had an associated AKI billing code.[24] Rates of AKI have also been defined in certain hospital settings, such as after major surgery. Estimates using Veterans Affairs data have suggested that the 7-day incidence of AKI after major surgery ranges from 4.1% after ear, nose, and throat surgery to 18.7% after cardiac surgery (Fig. 19.20).[186] Causes of AKI are rarely assessed in large studies, but some studies have suggested that among hospitalized patients, approximately two-thirds of AKI episodes were community-acquired and one-third were hospital-acquired.[187]

OUTCOMES

The study of AKI has been prioritized in both clinical and basic science research because of the recognition that outcomes, including development and progression of CKD, are worse among patients who experience even a small increase in serum creatinine level compared with those with stable values. In addition, studies have suggest that oliguria, even when not accompanied by a rise in serum creatinine level sufficient to meet the AKI criterion, may confer higher risk.[181,188] The incidence of CKD, decline in eGFR, ESKD, and death are all higher in patients who experience AKI, with higher AKI stage conferring higher risk.[179,189–193] For example, in one U.S. study, the risk of developing stage G4 or G5 CKD was 28 times higher after surviving an episode of KRT-requiring AKI.[194] Even modest changes in serum creatinine level have been associated with higher costs and length of stay. One 2005 study has estimated that an increase in serum creatinine level of 0.5 mg/dL or higher was associated with 3.5 additional hospital days and $7500 in excess costs.[195] The absolute risk of adverse outcomes varies based on underlying characteristics and stage of AKI, but many of those with AKI severe enough to require KRT never recover kidney function fully.[190,194,196] Controversial is whether the adverse outcomes are caused by AKI or simply due to shared risk factors.[197,198] There are no trials that have definitively proven causation due to the lack of effective therapy for AKI prevention and the need for very large sample sizes and long-term follow-up.[199]

RISK FACTORS

Risk factors for AKI are similar to those for CKD: older age, male gender and, variably, black race.[53] Racial disparities in AKI may be confounded by differences in socioeconomic

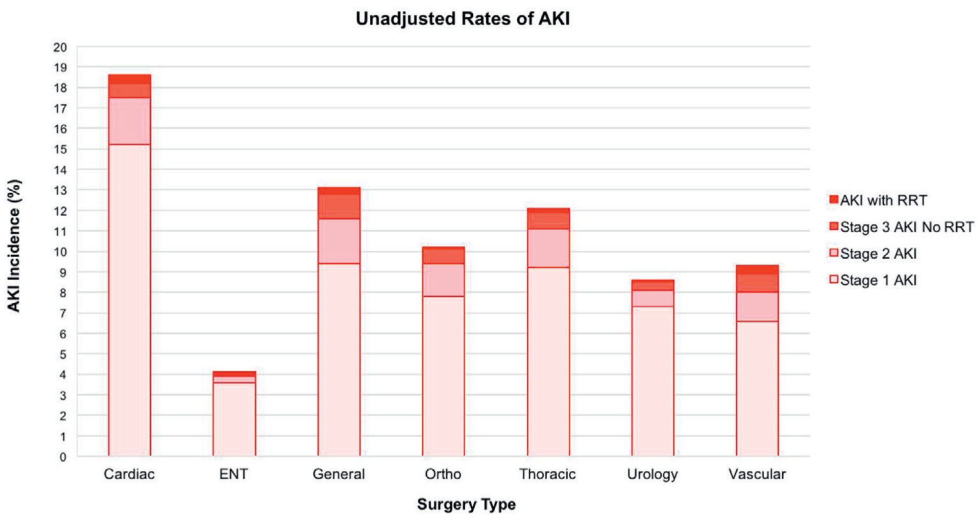


Fig. 19.20 Unadjusted rates of acute kidney injury, by surgery type and stage. *AKI,* Acute kidney injury; *ENT,* ear, nose, throat; *RRT,* renal replacement therapy. (Adapted from Grams ME, Sang Y, Coresh J, et al. Acute kidney injury after major surgery: a retrospective analysis of Veterans Health Administration data. *Am J Kidney Dis.* 2016;67[6]:872–880.)

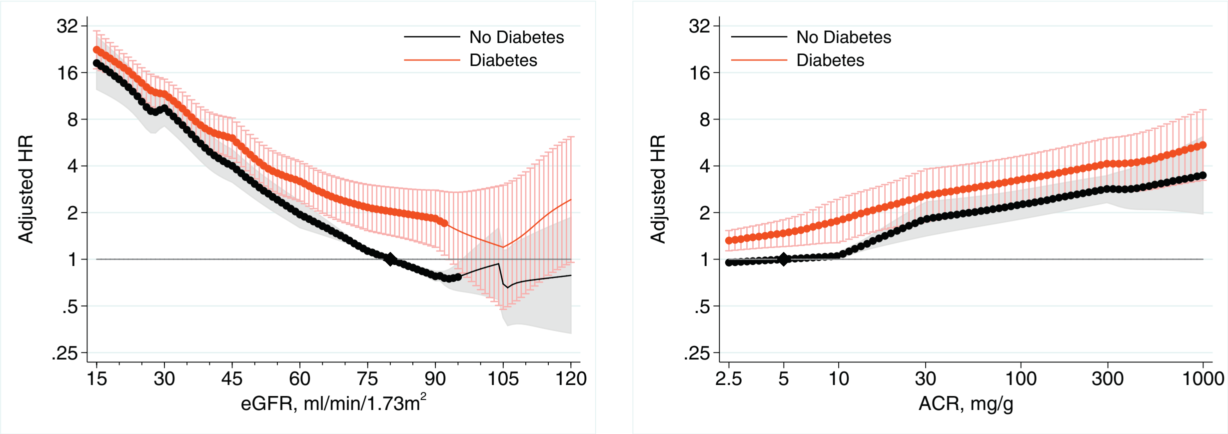


Fig. 19.21 Acute kidney injury risk by estimated glomerular filtration rate *(eGFR)* and urine albumin-to-creatinine ratio *(ACR),* stratified by diabetes status. Hazard ratios (*HRs;* with 95% CI [confidence interval]) of acute kidney injury, using common referent group without diabetes, according to eGFR (reference, 80 mL/min/1.73 m^2 without diabetes, left) and ACR (reference, 5 mg/g without diabetes; right) in individuals with and without diabetes in general population cohorts. (From James MT, Grams ME, Woodward M, et al. A meta-analysis of the association of estimated GFR, albuminuria, diabetes mellitus, and hypertension with acute kidney injury. *Am J Kidney Dis.* 2015;66:602–612.)

status—itself a risk factor for AKI—and, in a small study, were not explained by differences in kidney risk variants in APOL1.[200] Diabetes also confers an increased risk for AKI, with odds of AKI from 1.5- to 2.5-fold higher among persons with diabetes.[54,201] The strongest risk factors, however, appear to be lower eGFR and higher albuminuria, with each conferring a higher risk in a graded fashion, and little difference with the presence or absence of diabetes mellitus or hypertension (Fig. 19.21).

EVIDENCE GAPS AND LIMITATIONS OF THE LITERATURE

The literature of the epidemiology of kidney disease has important gaps:

1. Studies of the onset of disease are limited by the timing of and indication for ascertainment of kidney disease measures. Serum creatinine level is frequently measured during health encounters for acute and chronic illness. Markers of kidney damage are measured less frequently than serum creatinine level in clinical practice, and thus AKD and CKD are often underestimated. In research cohorts, measurements of serum creatinine levels are general not frequent enough to distinguish AKI, AKD, and CKD.
2. Factors other than GFR can affect serum concentrations of creatinine and cystatin C.
3. Many identified risk factors do not meet all the criteria for causality outlined by Bradford Hill.[7]
4. There are no uniform definitions for cause of kidney disease, and the clinical information necessary for ascertaining cause of disease is generally not available in research studies and clinical populations.
5. ESKD is defined by a therapy for which the indications for initiation are not uniform. Symptoms of kidney failure are poorly defined and ascertained, and the availability of KRT varies widely among countries, limiting inferences about kidney failure from ESKD studies.
6. Laboratory methods for the assessment of kidney measures can differ. Guidelines have improved comparability across studies by recommending standardization of assays for serum creatinine and cystatin C levels and routine reporting of eGFR, as well as measurement of urine albumin rather than total protein, but additional work is needed.

CONCLUSION

In conclusion, kidney disease is now recognized as a pressing global health concern. Over 10% of the world's population has CKD, and nearly 25% of the global hospitalized population experiences AKI. Enormous disparities exist in access to treatment, and millions of people are undertreated or untreated. Kidney disease epidemiology plays a critical role in providing evidence for the issuance of guideline definitions and recommendations, harmonization of research, development of trial endpoints, and success of global initiatives, such as those of the International Society of Nephrology, which seek to define global needs in CKD and eliminate preventable deaths from AKI by 2025.

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. A 78-year-old white man with diabetes, hypertension, and heart failure with preserved ejection fraction presents with a serum creatinine level of 1.2 mg/dL, eGFRcr of 58 mL/min/1.73 m^2, and a spot urine albumin-to-creatinine ratio of 1000 mg/g. According to the Kidney Disease Improving Global Outcomes (KDIGO) chronic kidney disease (CKD) staging and classification, he is
   a. No CKD, moderate risk
   b. CKD stage G3A2, moderate risk
   c. CKD stage G2A3, moderate risk
   d. CKD stage G3A3, very high risk
   Answer: d

2. A 24-year-old African American football player presents with a serum creatinine level of 1.7 mg/dL and eGFRcr of 55 mL/min/1.73 m^2. What additional test(s) might you order next to evaluate better whether he has kidney disease and to determine his prognosis?
   a. Spot urine albumin and creatinine
   b. Serum cystatin C
   c. Serum protein electrophoresis
   d. a and b
   e. All of the above
   Answer: d

3. Using the kidney failure risk equation, for the same age, gender, glomerular filtration rate, and level of albuminuria, a person in Europe would be predicted to have a ________ risk of end-stage kidney disease compared with a person in Canada.
   a. Higher
   b. Lower
   c. The same
   Answer: b

4. Which country has the highest prevalence of ESKD treated with dialysis?
   a. United States
   b. Thailand
   c. Taiwan
   d. Japan
   e. Belgium
   Answer: c